CEPHIONIX

NEURAL WINGS DIVISION

SYSTEM DESIGN PROPOSAL
AVIATION MANAGEMENT SYSTEM — AMS v2.0
Exclusively Designed and Developed for
HARYANA INSTITUTE OF CIVIL AVIATION (HICA)

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TABLE OF CONTENTS

Letter of Transmittal	Cover Letter
1. Executive Summary	Section 1
1.1 Purpose and Scope	
1.2 Critical Findings & Priority Gaps	
1.3 AMS Business Value Summary	
2. System Architecture	Section 2
2.1 Architectural Philosophy | 2.2 Technology Stack	
2.3 Deployment Model | 2.4 Multi-Tenancy	
2.5 Integration Architecture | 2.6 Security | 2.7 Disaster Recovery	
MODULE 1: User & Role Management (Onboarding Engine) — Flowchart + Full Spec	Module 1
MODULE 2: Flight Operations — Flowchart + Full Specification	Module 2
MODULE 3: Fleet Management — Flowchart + Full Specification	Module 3
MODULE 4: Fuel Management — Flowchart + Full Specification	Module 4
MODULE 5: Student Management — Flowchart + Full Specification	Module 5
MODULE 6: Instructor Management — Flowchart + Full Specification	Module 6
MODULE 7: Ground Training Management — Flowchart + Full Specification	Module 7
MODULE 8: AI Autonomous Rostering Engine (AIRE) — Flowchart + Full Spec	Module 8
MODULE 9: Meteorological Module — Flowchart + Full Specification	Module 9
MODULE 10: Safety Management System (SMS) — Flowchart + Full Specification	Module 10
MODULE 11: Legal & Compliance — Flowchart + Full Specification	Module 11
MODULE 12: Hostel Management — Flowchart + Full Specification	Module 12
MODULE 13: Finance & Accounting — Flowchart + Full Specification	Module 13
MODULE 14: Human Resources — Flowchart + Full Specification	Module 14
MODULE 15: DGCA Compliance Calendar — Flowchart + Full Specification	Module 15
MODULE 16: Flight Planning — Flowchart + Full Specification	Module 16
MODULE 17: Digital Logbook — Flowchart + Full Specification	Module 17
MODULE 18: Communication Engine — Flowchart + Full Specification	Module 18
MODULE 19: Reporting & Analytics — Flowchart + Full Specification	Module 19
3. Personnel Matrix — Current State & Gap Analysis	Section 4
4. Aircraft Management — Full Specification	Section 5
5. Fuel Management — Full Specification	Section 6
6. Instruments & Ground Equipment — Full Tracking Design	Section 7
7. DGCA Compliance Calendar — 35 Tracked Obligations	Section 11
8. Recommendations Table (REC-01 to REC-12)	Section 10

9. METEOROLOGICAL DEPARTMENT — FULL PLAN	Section 13
10. User Onboarding Flow (Staff | Aircraft | Branch | New FTO)	Section 12

Letter of Transmittal

Date: 2026

The Director

Haryana Institute of Civil Aviation (HICA)

Karnal (VIKR) | Pinjore, Haryana, India

Subject: Formal Submission — System Design Proposal for the HICA Aviation Management System (AMS v2.0)

Dear Sir / Ma'am,

On behalf of Neural Wings, the aviation technology division of Cephionix, we are honoured to present this comprehensive System Design Proposal for the HICA Aviation Management System (AMS) — a purpose-built, AI-augmented, cloud-native digital platform designed exclusively for your two operational branches at Karnal (VIKR) and Pinjore.

After conducting a thorough gap analysis of HICA's existing operational framework, our team has identified a set of critical, high-priority, and medium-priority compliance and operational deficiencies — including the absence of a LAME at both branches, the absence of a CFI at Pinjore, and the lack of a formal Safety Management System — that pose immediate regulatory and safety risk under the Aircraft Act 1934, Aircraft Rules 1937, and DGCA CAR Section 7, Series E.

The HICA AMS is not merely a software product — it is a strategic infrastructure transformation. Comprising 19 fully integrated modules, the system replaces every fragmented manual register, spreadsheet, and paper-based workflow with a single, DGCA-compliant digital source of truth. At its core is the AI Autonomous Rostering Engine (AIRE), a first-of-its-kind scheduling intelligence built specifically for Indian Flying Training Organisations, capable of generating an optimal daily sortie schedule in under 60 seconds while enforcing 100% of applicable DGCA regulatory limits.

The scope of this proposal covers the complete technical architecture, all 19 module specifications with process flowcharts, the personnel gap analysis, aircraft management and fuel management standards, the DGCA Compliance Calendar with 35 tracked obligations, and a commercial licensing model — all presented at the level of detail required for your technical review, your legal team, and your DGCA compliance auditors.

Neural Wings brings to this engagement a deep understanding of Indian aviation regulatory requirements, a technology stack built on proven enterprise-grade open-source components, and a commitment to delivering a system that not only meets DGCA expectations today but is architected to scale alongside HICA's ambition to expand from 2 to 10+ branches without a proportional increase in administrative overhead.

We respectfully request that the Honourable Director and the HICA management team review this proposal and extend us the opportunity to present a live demonstration at your earliest convenience. We are fully prepared to begin implementation upon your approval.

Yours faithfully,

Neural Wings Division

Cephionix

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1. EXECUTIVE SUMMARY

The HICA Aviation Management System (AMS) is an integrated, cloud-native, AI-augmented software platform purpose-built for DGCA-approved Flying Training Organisations (FTOs) operating in India. This document constitutes the authoritative System Design Specification (SDS) for HICA's two operational branches: Karnal (VIKR) and Pinjore (near Chandigarh).

The AMS replaces all fragmented manual registers, spreadsheets, and paper-based workflows with a single source of truth that enforces regulatory compliance, eliminates human error, and drives operational efficiency across every function — flight operations, fleet maintenance, fuel management, student training, personnel management, safety, finance, and legal compliance.

1.1 Purpose and Scope

This specification covers the exhaustive design of 19 integrated modules, an AI Autonomous Rostering Engine, a DGCA Compliance Calendar, and a Commercial Licensing Model. It addresses all current operational gaps at both HICA branches and provides a compliance-first blueprint that satisfies all obligations under the following regulatory instruments: DGCA CAR Section 7 Series E Parts I & II; Aircraft Act 1934 and Aircraft Rules 1937; DGCA CAR for Aircraft Maintenance; Aerodrome Standards (DGCA CAR Section 4); and ICAO Annexes 1, 6, and 14.

1.2 Critical Findings & Priority Gaps

A gap analysis of current HICA operations reveals the following non-compliant conditions that pose immediate regulatory and safety risk:

Gap	Branch	Regulatory Basis	Risk Level
No LAME (Cat B1) on strength	Both	Aircraft Rules 1937, Rule 62	CRITICAL
No CFI at Pinjore	Pinjore	CAR-FTO Para 3.1	CRITICAL
No Safety Officer / SMS Focal	Both	DGCA CAP 8100	HIGH
No Meteorological Officer or Department	Both	CAR Section 7, Series E	HIGH
No Ground Instructor at Karnal	Karnal	CAR-FTO Syllabus Requirements	HIGH
No AME at either branch	Both	Aircraft Act 1934	CRITICAL

1.3 AMS Business Value Summary

- Reduces daily scheduling time from ~2 hours to <5 minutes via the AI Rostering Engine
- Eliminates ~95% of manual DGCA compliance tracking effort via automated calendar and alerts
- Provides 100% digital audit trail for all flight, maintenance, and training records
- Enables HICA to scale from 2 to 10+ branches without proportional administrative overhead
- White-label SaaS model creates a secondary revenue stream by licensing AMS to other FTOs in India

2. SYSTEM ARCHITECTURE

2.1 Architectural Philosophy

The AMS is designed on a cloud-native, micro-services architecture following a Domain-Driven Design (DDD) approach. Each functional domain (Flight Ops, Fleet, Fuel, Students, etc.) is an independently deployable service with its own data store, communicating via a governed internal API bus. This architecture ensures that a failure or update in one domain does not cascade to others.

2.2 Technology Stack

Layer	Technology	Rationale
Frontend Web	React 18 + TypeScript + TailwindCSS	Component-based, type-safe, responsive UI
Mobile App	React Native (iOS & Android)	Cross-platform, offline-capable via SQLite sync
Backend API	Node.js (NestJS) + GraphQL + REST	Scalable, self-documenting, typed schemas
AI/ML Engine	Python (FastAPI) + TensorFlow/ PyTorch	Rostering, anomaly detection, NLP for reporting
Primary Database	PostgreSQL 15 (Multi-tenant with Row-Level Security)	ACID compliance, JSON support, audit triggers
Time-Series DB	TimescaleDB (PostgreSQL extension)	Aircraft metrics, fuel trends, weather data
Cache Layer	Redis 7 (Cluster Mode)	Session management, real-time pub/sub, job queues
Search Engine	Elasticsearch 8	Full-text search across all entities
Object Storage	AWS S3 / compatible (MinIO on-prem)	Documents, logbooks, photos, AFMs
Message Queue	Apache Kafka	Event streaming between microservices
Infrastructure	Kubernetes (EKS) + Helm Charts	Auto-scaling, self-healing, rolling deployments
CI/CD	GitHub Actions + ArgoCD	Automated testing, blue-green deployments
Monitoring	Prometheus + Grafana + PagerDuty	Real-time alerting, SLA tracking
Security	OAuth 2.0 + JWT + mTLS + Vault	Zero-trust inter-service, secrets management
Audit	Immutable audit log (WORM S3 + Blockchain hash)	Tamper-proof regulatory records

2.3 Deployment Model

2.3.1 Cloud (Primary)

Primary deployment on AWS Mumbai region (ap-south-1) for data residency compliance with Indian IT Act, 2000. Multi-AZ setup ensures 99.9% uptime SLA. All data encrypted at rest (AES-256) and in transit (TLS 1.3).

2.3.2 On-Premises Option

For clients requiring complete air-gap (e.g., defense-linked FTOs), a containerized on-premises deployment via Kubernetes on bare-metal servers is available. Synchronizes with cloud for updates when connectivity is available.

2.3.3 Offline Capability

Mobile app supports full offline operation for critical functions: pre-flight checks, snag logging, sortie start/end recording. Data syncs automatically when connectivity is restored, with conflict resolution managed by the CRDT (Conflict-free Replicated Data Type) engine.

2.4 Multi-Tenancy & Data Isolation

The AMS uses PostgreSQL Row-Level Security (RLS) with a tenant_id column on every table. A middleware interceptor injects the tenant_id into every query. Tenants (FTOs) cannot access each other's data under any circumstances. Branch-level scoping is an additional layer within each tenant.

2.5 Integration Architecture

External System	Integration Method	Data Exchanged
Garmin Pilot	REST API / Export parsing	Flight plans, track logs, fuel uplift
DGCA e-Services Portal	API + Scraper (where API unavailable)	DGCA SPL status, medical expiry, NOTAMs
IMD / AWC India	SOAP/REST API	METAR, TAF, SIGMET, winds aloft
NOTAM Service (AAI)	AFTN/AMHS polling	NOTAMs for VIKR, Chandigarh, Patiala
GST Portal (GSTN)	RESTful API	E-invoicing, GST filing
Banking / UPI	Razorpay / NEFT API	Fee collection, vendor payments
Biometric Devices	SDK integration (Mantra/Morpho)	Attendance, access control
ADS-B Receiver (OpenSky)	WebSocket API	Live aircraft position tracking
CAPA / Aviationweek DB	Licensed data feed	AD compliance auto-check
WhatsApp Business API	Meta Graph API	Automated notifications

External System	Integration Method	Data Exchanged
SMS Gateway	Twilio / MSG91	OTP, schedule alerts, emergency notifications

2.6 Security Architecture

- Authentication: OAuth 2.0 with PKCE, MFA mandatory for all roles above Student
- Authorization: RBAC with 47 granular permission bits organized into role templates (see Module 19)
- Data Encryption: AES-256 at rest; TLS 1.3 in transit; field-level encryption for Aadhaar/PAN
- API Security: Rate limiting (per-user, per-IP), OWASP Top-10 WAF rules, API key rotation
- Audit Trail: Every create/update/delete operation logged with user, timestamp, IP, old/new values; stored in WORM-compliant S3 with SHA-256 hash chain for tamper detection
- Penetration Testing: Mandatory annual third-party pentest; quarterly automated DAST scans

2.7 Disaster Recovery & Business Continuity

- RTO (Recovery Time Objective): < 4 hours
- RPO (Recovery Point Objective): < 1 hour
- Daily automated backups to S3 with 90-day retention; weekly backups retained 1 year
- Cross-region replication to AWS Singapore (ap-southeast-1) for geographic DR
- Failover: Automatic Route53 health-check based failover to standby region
- BCP Testing: Quarterly DR drills with documented results

MODULE 1 — User & Role Management

User & Role Management

START: New User / Organisation Onboarding Initiated

▼

RBAC Assignment: 14 Roles Defined — 47 Granular Permission Bits

(Super Admin | Director | CFI | DCFI | FI | LAME | Safety Officer | Student ...)

▼

Onboarding Wizard (8 Steps):

Organisation Setup → Branch → Aircraft → Personnel → Students → Config → Integration → Go-Live

▼

MFA Mandatory for All Roles Above Student

OAuth 2.0 + PKCE | AES-256 Encryption | WORM Audit Trail

▼

◆ **Go-Live Self-Check: All Critical Items Populated?**

▼

Gaps Flagged to Super Admin → Resolution Guided Step-by-Step

▼

System Live: All 19 Modules Active | Compliance Calendar Populated

Target Onboarding Time: < 48 Hours for Standard 2-Aircraft FTO

▼

END: HICA AMS Fully Operational — Audit-Ready from Day One

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MODULE 1 | USER & ROLE MANAGEMENT (ONBOARDING ENGINE)

USER & ROLE MANAGEMENT (ONBOARDING ENGINE)

1.1 Role-Based Access Control

Role	Modules Accessible	Write Permissions
Super Admin (HICA IT)	All modules, all branches, system config	Full
Director	All modules (own branch by default; cross-branch on request)	Read all; Write: HR, Finance, Compliance
CFI	Flight Ops, Fleet, Fuel, Student, Instructor, Met, SMS, Compliance	Full for branch
DCFI	Flight Ops, Fleet, Student, Instructor, Met	Full except staff management
Flight Instructor (FI)	Flight Ops (own sorties), Student, Digital Logbook, Snag raise	Own sorties only
CGI	Ground Training, Student (ground progress), Compliance Calendar, Study Materials	Ground training records
GI	Ground Training (own classes), Study Materials	Own class records
LAME/AME	Fleet (full), Snag (full), Parts Inventory, Fuel (quality tests)	Maintenance records only
Safety Officer	SMS (full), Compliance Calendar, Incident Reports, ERP	SMS and safety records
Accounts Officer	Finance (full), Procurement, Payroll	Finance records
Student Coordinator	Student (non-financial), Hostel, Ground Training (view)	Student admin records
Student	Own logbook (view), Own schedule, Own progress, Own documents	Own profile fields only
Hostel Warden	Hostel Module (full)	Hostel records
Storekeeper	Parts Inventory (full), Fuel (stock view)	Inventory records

1.2 Onboarding Wizard — New FTO

1. Step 1 — Organization Setup: Name, DGCA FTO number, GSTIN, registered address, logo, contact
2. Step 2 — Branch Setup: For each branch: name, ICAO code, aerodrome details, GPS coordinates
3. Step 3 — Aircraft Import: Wizard guides AME/CFI through adding each aircraft (C152/C172 template available)
4. Step 4 — Personnel Import: Bulk upload from Excel template or individual add via role wizard
5. Step 5 — Student Import: Bulk upload or individual addition

6. Step 6 — Configuration: Set DGCA compliance calendar dates, fuel reorder levels, duty time limits
7. Step 7 — Integration Setup: Connect Garmin Pilot, biometric devices, WhatsApp, SMS gateway
8. Step 8 — Go-Live Verification: System self-check verifies all critical items are populated; flags gaps

The entire onboarding wizard is designed for a target completion time of under 48 hours for a standard 2-aircraft, 2-instructor FTO.

1.3 Security and MFA

MFA is mandatory for all roles above Student. Authentication is implemented via OAuth 2.0 with PKCE. All data is encrypted with AES-256 at rest and TLS 1.3 in transit. Field-level encryption protects Aadhaar and PAN numbers. The complete audit log — every create, update, and delete operation with user identity, timestamp, IP address, and old/new values — is stored in WORM-compliant S3 with a SHA-256 hash chain for tamper detection. Annual third-party penetration testing and quarterly automated DAST scans are mandatory.

MODULE 2 — FLIGHT OPERATIONS

Flight Operations

START: CFI Creates Daily Flying Programme

▼

Go / No-Go Engine: 15+ Parameter Check

(Aircraft, Student, Instructor, Weather, Fuel, NOTAMs, W&B)

▼

◆ **All Parameters PASS?**

▼

Sortie Authorized — Dispatch Confirmed

ADS-B Tracking Activated | Overdue Timer Started

▼

In-Flight Monitoring: Position, Overdue Alerts, Emergency Protocol

▼

Post-Flight: Block Times, Fuel, Exercise Grades, Snag Logging

▼

Digital Logbook Auto-Populated → Instructor Digital Sign-Off

▼

SORTIE COMPLETE — RECORDS LOCKED & ARCHIVED

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MODULE 2 | FLIGHT OPERATIONS

FLIGHT OPERATIONS

2.1 Daily Flight Planning

The Flight Ops module is the operational nerve-centre of the AMS. Its primary function is to manage the complete lifecycle of every sortie from initial authorization through execution to post-flight debrief and record closure.

- Daily Ops Plan: CFI/DCFI creates the day's flying programme specifying: exercise per student, aircraft assignment, instructor assignment, tentative ETD/ETA, sortie type (Dual/Solo/Check/Nav)
- Dependency Check: System automatically validates all pre-flight dependencies before a sortie can be authorized (see 1.2)
- ATC Slot Management: Integration with ATC slot system if applicable; NOTAM auto-overlay on route
- Weather Hold: Automatic suspension of day's flying plan if wx falls below VFR minima with notification to all affected parties

2.2 Pre-Flight Authorization (Go / No-Go Engine)

The Go/No-Go engine is a rules-based AI module that evaluates 15+ parameters simultaneously before issuing dispatch authority:

Parameter	Source Module	Failure Action
Aircraft C of A valid	Fleet Module	Block dispatch; notify CFI and AME
Aircraft not U/S	Fleet / Snag Module	Block; show open snag detail
Aircraft 50/100/Annual not overdue	Fleet Module	Block; show hours remaining / overdue
Fuel sufficient for sortie + reserves	Fuel Module	Block; trigger fuel uplift request
Student medical valid (Class 2)	Student Module	Block; notify student and CFI
Student SPL valid	Student Module	Block; notify student
Instructor FI Rating valid	Instructor Module	Block; notify CFI
Instructor duty time within limits	Instructor Module	Warn at 80%; Block at 100%
Weather above VFR minima (WX Module)	Met Module	Block if below; AMBER if marginal
NOTAMs checked and acknowledged	Met/Ops Module	Block until acknowledged
W&B calculation within limits	Flight Planning Sub-module	Block if out of CG/weight limits
Performance calculation (T/O, Landing)	Flight Planning Sub-module	Warn if limiting; hard block if exceedance
ELT battery not expired	Fleet Module	Warn; non-blocking unless >30 days overdue

Parameter	Source Module	Failure Action
Student prerequisite exercise completed	Student Module	Warn; requires CFI override for solo/ check
ATC clearance (if required)	Comms Module	Non-blocking; alerts dispatcher

The Go/No-Go result is logged with timestamp, all parameter states, and the authorizing officer's digital signature. This constitutes the legal dispatch record.

2.3 Weight & Balance (W&B) Sub-Module

Each aircraft has a stored W&B template with: OEW, arm positions for each seat/ baggage bay, fuel arm. The CFI/DCFI enters: pilot weight, student weight, baggage weight, fuel quantity. The system calculates gross weight and CG location, plots on the W&B envelope diagram, and indicates GO/NO-GO. All results stored per sortie.

2.4 Sortie Dispatch and In-Flight Monitoring

- Dispatcher confirms engine start time (ETD)
- System activates overdue alert timer: EOBT + planned duration + 30 min buffer
- At EOBT + planned duration: AMBER alert to dispatcher and CFI
- At EOBT + planned duration + 30 min: RED OVERDUE alert to CFI, DCFI, and Emergency Contact; initiates Emergency Response Protocol
- ADS-B position tracking (if ADS-B Out installed): live track on map, geo-fence alerts

2.5 Post-Flight and Debrief

- Instructor enters: block-off, T/O, landing, block-on times; fuel uplifted; route flown
- Exercise grades entered per DGCA syllabus item (1–4 scale)
- Remarks for debrief recorded (voice-to-text AI transcription option)
- Snag creation: any defect observed triggers a snag record automatically linked to aircraft and sortie
- Student Digital Logbook auto-populated with verified sortie data; instructor digital sign-off
- Auto-calculation: student cumulative hours, dual hours, solo hours, XC hours, night hours

2.6 Emergency Management

The AMS contains a dedicated Emergency Response sub-module accessible from any screen via a red 'EMERGENCY' button. Activating it:

- Displays full Emergency Response Plan (ERP) for the relevant aerodrome
- Alerts all designated emergency personnel via SMS and push notification simultaneously
- Opens a live communication log for recording all emergency-related actions with timestamps
- Provides ATC frequency, nearest hospital, ambulance contact, fire brigade contact

- Auto-generates an ICAO Accident/Incident initial notification form

MODULE 3 — FEET MANAGEMENT

Fleet Management

START: Aircraft Added / Existing Aircraft Updated

▼

Aircraft Master Record: Reg, Serial, C of A, Insurance, AFM

▼

Continuous Tracking: Airframe | Engine | Propeller | Avionics Hours

▼

◆ **Due-Date Alert Triggered? (50hr / 100hr / Annual / AD)**

▼

Auto-Alert: CFI + AME Notified → Work Order Created in Maintenance Module

▼

Snag Raised → LAME Assigned → Rectification → QC Check

▼

◆ **Snag Rectified? CRS Issued by LAME?**

▼

Aircraft Returned to Serviceability → Go/No-Go Engine Updated

▼

END: Fleet Record Updated — Analytics Dashboard Refreshed

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MODULE 3 | FLEET MANAGEMENT

FLEET MANAGEMENT

3.1 Aircraft Master Record

The Fleet Management module provides exhaustive lifecycle tracking for every aircraft on HICA's register. Each aircraft has a master record containing all identity and document fields: Registration Mark (VT-XXX format validated against DGCA aircraft register API), manufacturer and model, serial number, year of manufacture, C of A (with 90/60/30-day advance alerts), C of R, Insurance, AFM/POH, MEL, and digital Journey Log and Technical Log equivalents.

3.2 Airframe, Engine, and Propeller Tracking

TTAF is auto-updated from sortie records. Hours since last 50-hour, 100-hour, and Annual inspection are continuously calculated, with automatic work order creation when under 10 hours remain. Engine tracking covers TSN, TSOH, TBO, oil change records, compression test results, and magneto timing. Propeller tracking includes TSN, TSOH, TBO, strike history — any ground strike triggers an immediate inspection requirement — and nick/damage records.

3.3 Airworthiness Directive Compliance

The AD compliance sub-module queries the FAA AD database and DGCA AD database using aircraft, engine, and propeller serial numbers. For each applicable AD, the system tracks: AD number, applicability confirmation, compliance time (one-time, recurring, on-condition, terminating), TTAF at last compliance, next compliance due (calculated automatically), and the LAME's sign-off. DGCA-issued ADs override FAA ADs where applicable.

3.4 Snag Management Workflow

The snag management workflow is a complete digital replacement for the paper snag book. A pilot or AME raises a Snag Record; the CFI reviews and classifies severity (AOG removes the aircraft from dispatch immediately); the snag is assigned to the LAME with a target completion date; the AME opens a Work Order with AMM reference, parts used (linked to inventory for stock deduction), man-hours, and photos; a quality check is performed for complex snags; and the LAME issues a Certificate of Release to Service digitally, returning the aircraft to serviceability. Recurring snags automatically notify the Safety Officer for trend analysis.

3.5 Avionics, Safety Equipment, and Parts Inventory

Every instrument — from the ASI and altimeter to the Garmin GTN GPS and transponder — is tracked individually with serial number, calibration dates, and next-due intervals. Safety equipment (fire extinguisher, first aid kit, seat belts, ELT) is tracked per aircraft. A Parts & Consumables inventory module tracks lubricants, filters, spark plugs, tyres, brake pads, batteries, and consumables — auto-generating Purchase Requisitions when stock falls to or below the reorder point.

MODULE 4 — FUEL MANAGEMENT

Fuel Management

START: Fuel Stock Drops Below Reorder Level

▼

Auto-PR Generated → Procurement Officer + CFI Notified

▼

PR Approved → PO Issued to Supplier (IOC/BPCL)

▼

Fuel Tanker Arrives → Delivery Logged (Qty, Batch, Temp, Density)

▼

5-Point Quality Acceptance Test:

Color | Clarity | Water | Density | Filter Condition

▼

◆ **All QAT Tests PASS? (AVGAS 100LL Verified?)**

▼

Fuel Added to Storage — Stock Level Updated — Batch Traced

▼

Per-Sortie Uplift Recorded → Consumption vs. Plan Variance Checked

▼

END: Monthly Fuel Audit — Reconciliation Report Generated

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MODULE 4 | FUEL MANAGEMENT

FUEL MANAGEMENT

4.1 AVGAS 100LL Enforcement

The AMS enforces AVGAS 100LL as the sole authorised fuel for all HICA aircraft. Any attempt to log or procure Jet-A, AVTUR, or MOGAS triggers a system-level CRITICAL ERROR with visual and audio alarm, blocking the action until an authorised officer explicitly acknowledges and records a documented reason. MOGAS use would require an STC which HICA aircraft do not hold.

4.2 Procurement Workflow

When stock falls below the reorder level, the system automatically creates a draft Purchase Requisition and alerts the Procurement Officer and CFI. Upon CFI/Director approval, a formal PO is issued to the fuel supplier (IOC Aviation/BPCL). On delivery, the storekeeper logs: delivery date and time, supplier name, vehicle registration, driver name, delivery note number, batch/lot number, quantity delivered in litres, temperature at delivery, and observed density (kg/L).

4.3 Quality Acceptance Testing

Before any fuel is accepted into storage, five mandatory quality acceptance tests must be recorded as PASS: Visual Color check (AVGAS 100LL is dyed blue — cloudiness or colour deviation triggers quarantine); Visual Clarity check (no visible solids or particulate); Water Detection test (using Millipore detector or AquaGlo, ensuring no free water and dissolved water below 15ppm); Density Check using a calibrated hydrometer (0.695–0.721 kg/L at 15°C per ASTM D6615); and Filter Condition inspection of the inline filter element. The system will not move fuel from 'Delivered — Pending QAT' status to 'Available Stock' until all five tests are recorded as PASS.

4.4 Storage Tracking, Per-Sortie Checks, and Monthly Audit

Storage tracking covers fixed underground tanks (with gauge calibration, integrity test, and bonding continuity dates), mobile bowsers (with flowmeter calibration, filter change log, and bonding reel condition), and drums (with batch traceability and storage condition records). Monthly compliance inspections — covering bonding cable continuity, fire extinguisher availability, spill kit, signage, and bund wall integrity — are mandatory Compliance Calendar items. Per sortie, pilots record sump drain checks, fuel cap security, quantity confirmed, and uplift quantity; actual vs. planned consumption variance triggers AMBER or RED alerts. The monthly automated fuel audit reconciles all uplifts, sortie usage, and physical stock measurements.

MODULE 5 — STUDENT MANAGEMENT

Student Management

START: New Student Enrolled — Comprehensive Profile Created

▼

DGCA Documents Verified: SPL | Class 2 Medical | RTF Licence

▼

Fee Structure Assigned → First Payment Collected → Digital Receipt

▼

Training Progress Tracker Initialised: 69 DGCA CPL Exercises

▼

Ground Training Enrolment: 7 CPL Theory Subjects Tracked

▼

◆ **Exercise Sign-Off Granted by Instructor?**

▼

Progress Advanced → Digital Logbook Updated → AI Trajectory Score

▼

◆ **All 69 Exercises + Theory Subjects Complete?**

▼

END: DGCA CPL Exam Eligibility Certified — Student Record Archived

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MODULE 5 | STUDENT MANAGEMENT

STUDENT MANAGEMENT

5.1 Student Profile

Each student has a comprehensive profile covering: personal data (full name, DOB, gender, Aadhaar, PAN, passport, blood group, emergency contact); academic background (10+2 marks in Physics and Math); DGCA credentials (SPL number, issue date, expiry, Class 2 medical expiry, AME name, RTF licence); financial data (course fee agreed, fee paid, balance, receipt numbers); encrypted bank details for scholarship and refund processing; and a digitised Documents Vault.

Data Category	Fields
Personal	Full name, DOB, gender, Aadhaar No., PAN, passport (if any), nationality, photo, contact, emergency contact, blood group
Academic Background	10+2 marks (Physics, Math), degree (if any), English proficiency test scores
DGCA Credentials	SPL number, issue date, expiry, class 2 medical expiry, medical examiner name, radio telephony license
Address	Permanent address, present address (hostel room number for residents)
Financial	Course fee agreed, fee paid, fee balance, payment history, fee receipt numbers
Bank Details	Bank, IFSC, account (encrypted) — for scholarship/refund processing
Documents Vault	Digitized copies of all certificates, IDs, medical reports, fee receipts

5.2 DGCA SPL & Medical Tracking

SPL numbers are recorded and verified against the DGCA eServices portal via API. Class 2 Medical expiry dates are stored with 60-day advance reminders, and students are automatically blocked from flying if their medical is expired. The Radio Telephony Licence is tracked separately as a prerequisite for solo cross-country flights. All seven CPL theory examination subjects are tracked with exam dates, scores, and pass/fail status.

5.3 Flight Training Progress — DGCA Syllabus Tracker

The module tracks all 69 standard DGCA CPL training exercises across five phases: Phase 1 (Basic Effects, Ex 1–10), Phase 2 (Circuits, Ex 11–20), Phase 3 (Advanced Flying, Ex 21–35), Phase 4 (Navigation, Ex 36–50), and Phase 5 (Pre-CPL Skills Test, Ex 51–69). For each exercise, the system records target completions, actual completions, instructor grades on a 1–4 scale, debrief comments, sign-off status, and the date on which the student is certified to advance to the next exercise.

5.4 Ground Training, Fee Management, and AI Trajectory

The ground training sub-tracker records class attendance (biometric sign-in option), test scores, pass/fail status, remedial sessions, and examiner sign-off for CPL theory examination eligibility. The fee management integration provides a real-time fee ledger with automated payment reminders and GST-compliant digital receipts. Fee defaulters are flagged, blocking flight authorisation and ground exam access pending CFI override. The AI trajectory engine flags students at risk of falling behind their CPL hour target based on their current completion rate and historical pace.

MODULE 6 — INSTRUCTOR MANAGEMENT

Instructor Management

START: Instructor Profile Created — Credentials Uploaded

▼

Licence & Ratings Verified: CPL/ATPL | FI Rating | Aircraft Endorsements

▼

Class 1 Medical Logged → Expiry Reminders Set (90 / 60 / 30 days)

▼

Daily Duty Time Accumulation Tracked vs. DGCA Limits

(8hr / 30hr / 100hr / 900hr)

▼

◆ **Approaching Duty Limit? (80% Warning Threshold)**

▼

WARN: Alert to CFI + Instructor → Sorties Blocked at 100%

▼

FI Rating Renewal Tracker: 7-day, 30-day, 90-day Automated Reminders

▼

◆ **FI Rating or Medical Expired?**

▼

END: Instructor Auto-Blocked from Dispatch — CFI Notified

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MODULE 6 | INSTRUCTOR MANAGEMENT

INSTRUCTOR MANAGEMENT

6.1 Instructor Profile

Data Category	Fields
Identity	Name, employee ID, photo, DOB, contact
License & Ratings	CPL/ATPL number, FI Rating number, FI Rating expiry, aircraft endorsements (C152, C172), instrument rating (if any), night rating
Medical	Class 1 medical certificate number, expiry, AME name
Flying Currency	Last 90-day currency per aircraft type; system tracks automatically from sortie records
Duty Time	Daily, monthly, and annual flight duty time tracked vs. DGCA limits
Performance	Student pass rates, average grades given, student feedback scores

6.2 DGCA Duty Time Limits Enforcement

The system enforces the following limits automatically (DGCA CAR Section 7, Series E, Part I):

- Maximum flight duty time: 8 hours in any 24-hour period
- Maximum flying hours: 30 hours in any 7-day period, 100 hours in any 28-day period
- Annual maximum: 900 hours per calendar year
- Minimum rest between duty periods: 12 consecutive hours

The system will WARN at 80% of any limit and BLOCK authorization when a limit is reached.

6.3 FI Rating Currency and Renewal

FI Rating is valid for 1 year. System issues a 90-day, 30-day, and 7-day automated reminder. At expiry, instructor is automatically removed from dispatch eligibility and CFI is notified. Renewal documentation checklist provided within the module.

MODULE 7 — GROUND TRAINING MANAGEMENT

Ground Training Management

START: CGI Builds Ground Training Curriculum — Topic Library Created

▼

7 CPL Subjects Scheduled: Air Regs | Nav | Met | Tech Gen | Tech Spec | RTF | Airlaw

▼

Class Scheduled in AI Rostering Engine → Attendance Marked Digitally

▼

Study Materials Uploaded: PDFs, Video Lectures (Version-Controlled)

▼

AI Mock Exam Generated: 5,000+ Question Bank → Personalised per Student

▼

◆ **Student Achieves Pass Mark in All Theory Subjects?**

▼

Instructor Signs Off Theory Completion → DGCA Exam Eligibility Updated

▼

END: Ground Training Complete — Student CPL Theory Record Finalised

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MODULE 7 | GROUND TRAINING MANAGEMENT

GROUND TRAINING MANAGEMENT

7.1 Curriculum Builder

The CGI can create and manage the ground training curriculum for all CPL subjects: Air Regulations, Air Navigation, Meteorology, Technical General, Technical Specific (Cessna 152/172), and Radio Telephony. Each subject has a topic library with lessons, learning objectives, and reference materials. The curriculum builder supports version control, ensuring students always receive the most current approved content.

7.2 Class Management

- Class scheduling integrated with the AI Rostering Engine (ground classes are a resource that competes with flying for instructor time)
- Digital attendance register: student sign-in via app; biometric option available
- Session recording upload capability (video lectures)
- Study Material Library: PDF/video upload, searchable, version-controlled

7.3 AI-Powered Mock Exam System

- Question bank with 5,000+ questions across all CPL subjects, tagged by topic, difficulty, and DGCA exam weighting
- AI generates unique mock exams for each student based on their weak areas identified from previous test performance
- Instant grading and performance analytics: subject-wise breakdown, weak topics, improvement trajectory
- Exam simulation mode: timed, invigilated, no review — mirrors actual DGCA ATPL/CPL exam conditions

MODULE 8 — AI AUTONOMOUS ROSTERING ENGINE

AI Autonomous Rostering Engine

START: 06:30 Daily Trigger (or Manual CFI Trigger)

▼

AIRE Ingests 25+ Live Parameters:

Weather | Aircraft | Instructor | Student | Fuel | NOTAMs | Ground Schedule

▼

STAGE 1 — Hard Constraint Engine (Google OR-Tools CP-SAT)

Eliminate All Illegal / Unsafe Pairings

▼

STAGE 2 — Genetic Algorithm Optimization

500 Populations × 200 Generations → Best Valid Schedule

▼

STAGE 3 — Reinforcement Learning Loop

Learn from Overrides & Outcomes → Weight Adjustment

▼

◆ **CFI Reviews & Approves Roster?**

▼

Communication Cascade: SMS + WhatsApp to All Rostered Parties

▼

END: Roster Published — Real-Time Re-Optimisation on Any Change

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MODULE 8: AI AUTONOMOUS ROSTERING ENGINE (AIRE)
FULL DESIGN SPECIFICATION — REVISED & ENHANCED

Document Reference: HICA-AMS-SDS-NEURAL-001
Classification: RESTRICTED — FOR INTERNAL USE ONLY
Applicable To: HICA Karnal (VIKR) | HICA Pinjore

1. OVERVIEW & DESIGN PHILOSOPHY

The AI Autonomous Rostering Engine (AIRE) is the most operationally critical and technically sophisticated component of the HICA AMS. It is not a static calendar or a simple booking tool. AIRE is a continuously running, self-adapting intelligence engine that manages two distinct but deeply linked scheduling horizons:

Horizon	Scope	Output	Trigger
MONTHLY ROSTER	Full calendar month, generated on the 25th of preceding month	Published duty calendar for all instructors, students, aircraft	Auto-generated monthly + manual CFI trigger
DAILY ROSTER	Operational sortie schedule for the current flying day	Time-blocked sortie assignments + communication cascade	Auto at 06:30 daily + any real-time event (snag, weather, sick-call)

AIRE operates on a foundational principle: every decision it makes must be explainable, auditable, and overridable. No AI output is ever silently applied — every schedule is presented to the CFI/DCFI for review before publication, and every override is logged with a mandatory reason.

CORE PROMISE	AIRE guarantees: (1) Zero hard-constraint violations in any published roster. (2) Automatic cascade communication on every schedule change — instructors and students are never left uninformed. (3) Continuous self-improvement from real-world outcomes.

2. SYSTEM ARCHITECTURE — FOUR LAYERS

AIRE is structured as four interdependent layers that execute in a defined sequence:

Layer	Component	Technology	Role
1 — DATA	Live Data Ingestion Bus	PostgreSQL RLS + Event Stream	Pulls real-time status from all 19 AMS modules continuously
2 — RULES	Hard Constraint Engine (CSP)	Google OR-Tools CP-SAT Solver	Eliminates all illegal or unsafe pairings. Cannot be bypassed.
3 — OPTIMIZE	Soft Optimization Engine (GA)	Genetic Algorithm — Population 500, 200 Generations	Finds the best valid schedule from thousands of possibilities

4 — LEARN	Reinforcement Learning Loop (RL)	Feedback-driven weight adjustment	Improves scheduling quality continuously from human overrides and real outcomes

3. TWO-HORIZON SCHEDULING DESIGN

3.1 Monthly Roster Generation

The monthly roster is the strategic planning layer. It gives instructors, students, and management a confirmed schedule one month in advance, enabling personal planning and ensuring DGCA duty time limits are respected across the full month.

3.1.1 Generation Trigger

- Auto-triggered on the 25th of every month at 18:00 for the following month
- CFI or Director can manually trigger a regeneration at any time
- A 'Draft Review' version is shared with CFI first; only 'Published' version goes to staff

3.1.2 Monthly Inputs

- Instructor leave applications (approved/pending) from HR Module
- Student examination schedules (DGCA written exams, check rides)
- Aircraft scheduled maintenance windows (from Fleet Module — 50hr, 100hr, Annual)
- Public holidays and aerodrome closure dates (pre-loaded compliance calendar)
- Instructor DGCA duty time balance (hours flown this month → limits remaining)
- Student training progress and monthly hour targets (from Student Module)
- Ground training class schedule requirements (from Ground Training Module)

3.1.3 Monthly Outputs

- Duty calendar per instructor: duty days, off days, leave days, training days
- Student flying day allocation: which days each student is rostered to fly
- Aircraft maintenance scheduling: which days each aircraft is in for service
- Ground training schedule: classroom days per batch
- DGCA duty time projection: forecast of instructor hours to confirm no limit will be breached

DGCA COMPLIANCE LOCK	The monthly roster is generated with a hard cap ensuring no instructor will exceed DGCA limits even if all rostered days are flown. The system reserves a 10% safety buffer below the 30-hour/7-day and 100-hour/28-day limits.

3.2 Daily Roster Generation

The daily roster is the operational execution layer. While the monthly roster sets the broad framework, the daily roster is a precision instrument that assigns specific aircraft, instructors, students, and time slots to each sortie — and adapts in real time to any change.

3.2.1 Generation Trigger — 06:30 Daily

Each flying day, AIRE executes a full daily roster generation at 06:30, as the aerodrome begins its opening sequence. This timing is deliberately chosen to balance two competing needs: incorporating the most current early-morning data (weather, overnight snags, instructor confirmations) while ensuring the roster is ready well before the first sortie departs. This ensures:

- The latest weather data (early-morning METAR and updated TAF) is incorporated before any sortie is committed
- Any early-morning snags (raised by the AME during the pre-flight inspection that begins at 06:00) are already reflected in the roster before the first student arrives
- Instructor availability confirmed via their app acknowledgment (required by 07:00)
- CFI receives the draft roster by 07:15 for review before the 07:30 operations briefing

3.2.2 Daily Inputs — 25+ Live Parameters

Category	Parameter	Source Module	Impact if Unavailable
WEATHER	Live METAR (Visibility, Wind, Ceiling, QNH)	Met Module / AAI Data Feed	All VFR flights suspended — Weather Hold declared
WEATHER	TAF (6-hour forecast)	Met Module	Late-opening slots re-evaluated
WEATHER	Active NOTAMs	Flight Planning Module	Affected route sorties suspended
WEATHER	Civil Twilight (Sunrise/ Sunset)	System Clock + Location	Night / early morning sorties adjusted
AIRCRAFT	Serviceability Status (Serviceable / AOG / Limited)	Fleet Module — CRS Status	Grounded aircraft removed from dispatch pool
AIRCRAFT	Hours until 50-hr / 100-hr inspection	Fleet Module	Aircraft near limit may be restricted to shorter sorties

AIRCRAFT	Fuel on board + Fuel stock at branch	Fuel Module	Sorties reduced if fuel insufficient
AIRCRAFT	Valid Certificate of Airworthiness + Insurance	Fleet Module — Document Vault	Aircraft blocked from dispatch if any document expired
INSTRUCTOR	On-duty / On-leave / Sick / Absent status	HR Module + App check-in	Instructor removed from pool; student cascade triggered
INSTRUCTOR	Remaining daily duty hours (DGCA limits)	Instructor Module	Limits block additional sortie assignments
INSTRUCTOR	FI Rating and Medical currency	Instructor Module	Expired instructor auto-removed from dispatch
INSTRUCTOR	Assigned students + current exercise	Student Module	Ensures correct pairing for syllabus continuity
STUDENT	Medical certificate (Class 2) validity	Student Module — DGCA API	Student blocked from flying if expired
STUDENT	Next required DGCA exercise	Student Module	Determines sortie type (Dual/Solo/ Nav/Check)
STUDENT	Training priority score (see 4.2)	AIRE Priority Engine	High-priority students scheduled first
STUDENT	Fee account status	Finance Module	Defaulters blocked pending CFI override
STUDENT	Availability confirmation	App check-in by 07:00	Absent students replaced from standby list
FUEL	AVGAS 100LL stock level vs. day's planned consumption	Fuel Module	Day's sortie count capped if fuel insufficient
MAINTENANCE	Open AOG snags	Fleet Module — Snag Register	Aircraft removed from pool immediately
OPERATIONS	Yesterday's planned vs. actual sorties	Flight Ops Module	Informs compensation scheduling for missed sorties
OPERATIONS	Ground training class schedule	Ground Training Module	Ensures no double-booking of instructor time
COMPLIANCE	Any compliance item at T-0 (expired today)	DGCA Compliance Calendar	Alert to CFI; affected operations paused if required

4. INTELLIGENT REAL-TIME DECISION SCENARIOS

This section documents, in precise detail, how AIRE responds to the most common real-world operational disruptions. These are not manual processes — they are fully automated decision chains executed by the system within seconds of a trigger event.

4.1 Scenario: Weather Opens Late

REAL-WORLD TRIGGER	At 06:30, the METAR shows visibility 2.5km (below 5km VFR training minimum) and a low cloud ceiling of 800ft. The TAF predicts improvement to 6km / ceiling 3,000ft by 09:00.

Step-by-Step AIRE Response:

1. AIRE reads the 06:30 METAR. Weather conditions fail VFR minima check. All planned morning sorties are flagged SUSPENDED — PENDING WEATHER.
2. AIRE reads the TAF. It detects the forecast improvement window (09:00 onwards). It does NOT cancel the day's flying — it defers and re-slots.
3. STUDENT PRIORITY RANKING: AIRE ranks all rostered students by their priority score (see Section 4.1.1). Students with the highest urgency get the limited afternoon slots. Lower-priority students are moved to 'Deferred — Flying Tomorrow (Priority)' status.
4. CFI is presented with the revised roster showing: TODAY'S HIGH-PRIORITY STUDENTS (who will fly when weather opens), and DEFERRED STUDENTS (who get priority compensation on the next available day).
5. **COMMUNICATION CASCADE (Auto-Sent after CFI approves):**
   - HIGH-PRIORITY STUDENTS → SMS + WhatsApp: "Flying today. Report to aerodrome by [time]. Weather expected to open at 09:00. Await confirmation before departing."
   - DEFERRED STUDENTS → SMS + WhatsApp: "Flying deferred today due to weather. You have been given PRIORITY status for tomorrow's roster. We will confirm your slot by [time] today."
   - ALL INSTRUCTORS → SMS + WhatsApp: "Weather delay today. Revised flying programme starts [time]. Please report by [time]."
   - CFI DASHBOARD → Real-time notification with revised sortie count and weather timeline
6. LIVE MONITORING: AIRE monitors METAR updates every 30 minutes. When weather meets minima, it auto-triggers a 'Weather Opening' notification to all rostered parties. If weather does not open by 12:00, full day cancellation is declared and compensation scheduling is applied to the next 3 days.

4.1.1 Student Priority Scoring System

When slots are limited, AIRE uses a calculated Priority Score (0–100) to rank students. Higher score = gets the available slot.

Priority Factor	Weight	Calculation Logic
Hours Behind Monthly Target	35%	Target monthly hours - actual hours flown. More behind = higher score.
Proximity to Milestone	25%	Within 2 hours of first solo, first XC, or skills test = maximum points
Days Since Last Flight	20%	Every day without flying adds points (currency risk after 28 days)
Phase of Training	10%	Students in Phase 4/5 (pre-CPL) get higher weight than Phase 1
Medical Expiry Urgency	10%	If medical expires within 30 days, student gets boosted priority to fly before expiry

4.2 Scenario: Instructor Calls In Sick / On Leave

REAL-WORLD TRIGGER	Instructor A sends a sick-leave message via app at 07:00. He was assigned to 3 students today — Student X (Dual Exercise 15), Student Y (Solo Circuit), Student Z (Nav Briefing).

Step-by-Step AIRE Response:

7. Instructor A's status is marked SICK via app. AIRE immediately removes Instructor A from today's dispatch pool.
8. REPLACEMENT LOGIC: AIRE checks all other available instructors for: (a) qualification to take each student's exercise, (b) remaining duty hours capacity, (c) current student load today.
9. If a qualified replacement is available: AIRE re-assigns students to the available instructor and re-generates the day's sortie time slots to accommodate the new pairings.
10. If NO qualified replacement is available: Affected students are marked DEFERRED with Priority status for tomorrow. CFI is notified with a gap alert.
11. **COMMUNICATION CASCADE (Auto-Sent immediately after CFI acknowledges):**
    - AFFECTED STUDENTS (if NO replacement found) → SMS + WhatsApp: "Dear [Name], your instructor [Instructor A] is unavailable today. Your flying session has been deferred. You DO NOT need to come to the aerodrome today. Your slot will be given PRIORITY on the next flying day. We apologise for the inconvenience."
    - AFFECTED STUDENTS (if replacement FOUND) → SMS + WhatsApp: "Dear [Name], your instructor today will be [Instructor B] instead of [Instructor A]. Your time slot is [new time]. Please report as planned."
    - CFI DASHBOARD → Alert showing: Instructor sick, students affected, replacement status, revised roster
    - HR MODULE → Sick leave auto-created for Instructor A (pending documentation)

4.3 Scenario: Mid-Day Aircraft Snag (AOG)

REAL-WORLD TRIGGER	Aircraft VT-HCA completes its 09:30 sortie. The pilot reports a rough engine on landing. AME inspects and raises an AOG snag at 11:00. Two afternoon sorties on VT-HCA are now impossible.

Step-by-Step AIRE Response:

12. AME raises AOG snag in Fleet Module. AIRE receives event trigger within 30 seconds. VT-HCA is immediately removed from dispatch availability.

13. AIRCRAFT SWAP LOGIC: AIRE checks if another serviceable aircraft can cover the affected sorties (considering fuel, hours, availability). If yes, it re-assigns and re-slots.
14. If aircraft swap is not possible: Affected students are informed, marked DEFERRED with Priority, and their compensation is scheduled on the next available flying day.
15. **COMMUNICATION CASCADE:**
    - AFFECTED STUDENTS → SMS + WhatsApp: "Dear [Name], your afternoon sortie has been cancelled due to an aircraft technical issue. You have been given PRIORITY for your next flying day. No action needed — we will confirm your slot."
    - INSTRUCTOR → SMS + WhatsApp: "Your [time] sortie with [Student] has been cancelled (aircraft AOG). Please stand by for revised duties or contact the CFI."
    - CFI + AME → Push notification with snag details and revised sortie count

4.4 Scenario: Student Fails to Check In

REAL-WORLD TRIGGER	Student Y was rostered to fly at 09:00. By 08:45, Student Y has not confirmed attendance via the app and is not present at the aerodrome.

Step-by-Step AIRE Response:

16. At T-15 minutes (08:45), AIRE detects no attendance confirmation for Student Y. Auto-alert sent: "[Name], your flight is at 09:00. Please confirm attendance or contact the aerodrome immediately."
17. At T-0 (09:00), if no response: AIRE checks the STANDBY LIST — students not originally rostered today but available and eligible. If a standby student exists, AIRE proposes a substitution to CFI.
18. CFI approves substitution with one tap. AIRE re-assigns the instructor and aircraft to the standby student and logs the swap.
19. Student Y is marked ABSENT-NO-SHOW. The incident is flagged in their Student Profile for CFI review. No priority compensation is given for no-shows (unlike weather/snag deferrals).

4.5 Scenario: Deferred Student Compensation Scheduling

Whenever a student's sortie is deferred for a reason outside their control (weather, snag, instructor sick), AIRE automatically manages the compensation:

- Student is flagged with a COMPENSATION DUE status in their profile
- Their priority score is elevated for the next available flying day
- The monthly target hours are automatically recalculated to reflect the lost day

- If multiple compensation days accumulate, AIRE may propose a Saturday or extended day flying session to recover hours, subject to CFI approval
- Compensation scheduling is shown on a dedicated 'Deferred Hours Recovery' dashboard for the CFI

FAIRNESS GUARANTEE	AIRE tracks every deferred sortie and ensures that no student is systematically disadvantaged due to recurring weather or equipment issues. The priority system ensures equitable recovery across all students.

5. ALGORITHM DESIGN — THREE STAGES

5.1 Stage 1: Hard Constraint Engine (CSP)

The first stage is a binary filter. Every possible student-instructor-aircraft-time combination is evaluated against the hard constraints listed below. Any combination that violates even one hard constraint is immediately discarded. This stage uses Google OR-Tools CP-SAT Solver.

Hard Constraint	Check Logic	Outcome if Failed
Aircraft Serviceability	CRS status = Serviceable AND no open AOG snags	Aircraft removed from pool
Aircraft Documentation	C of A + Insurance + Radio License all valid today	Aircraft blocked until documents renewed
Aircraft Hours	Current TTAF < (next inspection due TTAF)	Aircraft restricted or grounded
Instructor Duty Time	Today's planned hours + existing duty < DGCA daily limit (8 hrs)	Instructor blocked from additional sorties
Instructor Weekly Limit	Week's total + today's planned ≤ 30 hours	Instructor blocked
Instructor Medical	Class 1 Medical expiry > today's date	Instructor removed from pool
Instructor FI Rating	FI Rating expiry > today's date	Instructor removed from pool
Student Medical	Class 2 Medical expiry > today's date	Student blocked from flying
Student Exercise Eligibility	Previous exercise signed off by instructor	Student cannot advance to next exercise
Weather VFR Minima	Visibility ≥ 5km AND Ceiling ≥ 1,500ft AGL	All VFR training suspended
Fuel Sufficiency	Available stock ≥ total planned consumption + 20% reserve	Sorties reduced
VFR Day Limits	ETD and ETA both within civil twilight	Night sorties blocked (unless night rating exercise)

5.2 Stage 2: Genetic Algorithm Optimization

With all hard constraints satisfied, Stage 2 finds the BEST valid schedule. The Genetic Algorithm runs 500 simulated schedules across 200 generations, evolving toward the highest composite fitness score.

Optimization Goal	Weight	Scoring Logic
Student Training Efficiency	40%	Priority score of each student scheduled × hours assigned
Instructor Workload Balance	20%	Standard deviation of duty hours across instructors (lower = better)
Aircraft Utilization	15%	Total aircraft flying hours as % of available daylight time
Compensation Hour Recovery	15%	Points awarded for scheduling students with COMPENSATION DUE status
Instructor-Student Continuity	10%	Bonus points when a student flies with their primary assigned instructor

5.3 Stage 3: Reinforcement Learning Loop

AIRE records every human override and every real-world outcome. Over time, it builds a model of what makes schedules work in the real world at HICA specifically — not just in theory.

- CFI overrides an AI-assigned pairing (e.g., moves Student X off Instructor A) → AIRE logs this pattern and progressively reduces the weight of that pairing in future schedules
- A sortie that was scheduled is cancelled due to student no-show → AIRE learns to check that student's historical attendance rate and adjust their standby probability
- Weather frequently delays flying until 10:00 in November → AIRE learns to generate schedules with later start times in November, reducing wasted check-in efforts
- Certain instructor-student pairings consistently achieve Grade 4 (highest) exercise scores → AIRE prefers these pairings for milestone sorties

The RL layer runs a planned vs. actual analysis every Monday morning, covering the previous week. Results are shown in the AI Insights dashboard (Module 18) for CFI review.

6. AUTOMATED COMMUNICATION ARCHITECTURE

AIRE has a built-in Communication Engine that ensures no roster change, cancellation, or rescheduling event is ever communicated manually. Every event triggers a defined, role-specific message cascade.

Event	Recipients	Channel	Message Template
Weather Hold Declared	All rostered students + instructors	SMS + WhatsApp	"Today's flying is on WEATHER HOLD. [Situation]. We will update you by [time]."
Weather Opening Confirmed	High-priority students + instructors	SMS + WhatsApp	"Flying is GO for today. Report to aerodrome by [time]. Your sortie: [details]."
Full Day Cancellation	All rostered students + instructors	SMS + WhatsApp + App Push	"Flying CANCELLED today. You have been given PRIORITY status for [date]. No action needed."
Instructor Sick — No Replacement	Affected students only	SMS + WhatsApp	"Your instructor is unavailable today. DO NOT come to aerodrome. Priority slot confirmed for [date]."
Instructor Replaced	Affected students only	SMS + WhatsApp	"Your instructor today is [Name] (replacing [Name]). Slot unchanged: [time]."
Aircraft AOG — Sortie Cancelled	Affected student + instructor	SMS + WhatsApp	"Your sortie is cancelled (aircraft technical issue). Priority slot: [date]. Sorry for the inconvenience."
Student No-Show Alert	Student (T-15 min)	SMS + WhatsApp	"Your flight is at [time]. Please confirm attendance or call the aerodrome immediately."
Monthly Roster Published	All staff + students	Email + App Push	"Your roster for [Month Year] has been published. View it in your AMS app."
Daily Roster Published	All rostered parties	App Push	"Today's flying programme is confirmed. View your slot in the app."
Priority Slot Confirmed	Deferred student	SMS + WhatsApp	"Your deferred sortie has been rescheduled. NEW DATE: [date], TIME: [time]. Please confirm."

MESSAGE APPROVAL	All automated messages are pre-approved templates. The CFI approves the communication cascade with a single tap after reviewing the revised roster. No message is sent without CFI acknowledgment.

7. HUMAN OVERRIDE PROTOCOL

AIRE is a tool that assists the CFI — it never replaces their judgment. The override system is designed to be fast, logged, and fully auditable.

- Any authorized user (CFI / DCFI / Director) can modify any AI-assigned sortie, instructor pairing, or time slot
- Every override requires: Who changed it, What was the original assignment, What is the new assignment, Mandatory reason text (min 10 characters)

- After an override, AIRE immediately re-runs the optimizer for the remaining unconfirmed sorties around the locked change
- Hard constraint violations CANNOT be overridden through the schedule UI — they require resolution in the source module (e.g., LAME must issue CRS before aircraft appears available)
- Override patterns are fed into the RL learning pipeline weekly

Override Type	Who Can Override	System Response
Swap student between instructors	CFI / DCFI	Re-optimize remaining schedule around fixed swap
Change sortie time slot	CFI / DCFI	Check for conflicts; re-slot if conflict-free
Cancel a specific sortie	CFI / DCFI	Student marked DEFERRED-PRIORITY; communication triggered
Add an unrostered student	CFI only	Check student eligibility; add if constraints satisfied
Cancel full day's flying	CFI / Director	All students marked DEFERRED-PRIORITY; full cascade communication
Extend flying beyond planned hours	CFI only	DGCA duty time check; BLOCK if limit would be breached

8. PERFORMANCE REQUIREMENTS

Function	Performance Target	Notes
Monthly roster generation	< 5 minutes	Runs overnight; time is not critical
Daily roster initial generation (06:30)	< 60 seconds	Full 3-aircraft, 6-student, 3-instructor schedule
Re-optimization after single override	< 10 seconds	Real-time re-slot of remaining sorties
Re-optimization after major event (AOG / Weather)	< 30 seconds	Full re-plan triggered by module event
Communication cascade dispatch	< 60 seconds after CFI approval	SMS + WhatsApp via gateway API
Priority score recalculation	< 5 seconds	Triggered after every sortie completion or cancellation
RL model weekly update	< 30 minutes	Runs Sunday night; does not affect live operations

9. SUMMARY — WHAT MAKES HICA AIRE UNIQUE

Capability	AIRE (HICA AMS)	Typical Manual System
Monthly roster generation	Automated in minutes with DGCA compliance built-in	2-3 hours manual, error-prone
Daily roster adaptation	Real-time re-planning in <30 seconds	Manual phone calls, delays

Weather delay management	Auto-prioritization + student communication cascade	Manual CFI calls, students come unnecessarily
Instructor sick call	Auto-reassign or auto-defer + instant student notification	Students arrive and find no instructor
Compensation tracking	Every deferred sortie tracked; auto-priority on next day	Not tracked; students lose hours
DGCA compliance	Every roster pre-validated against all DGCA limits	Checked manually; errors common
Communication	Automated SMS/WhatsApp to every affected party	Manual calls, often missed
Learning	Improves with every week of operation	Stays as good (or bad) as the operator

BOTTOM LINE	AIRE transforms roster management from a 2-hour daily administrative burden into a 5-minute review-and-approve workflow for the CFI, while ensuring that every student, every day, knows exactly where they stand — and that no flying opportunity is ever wasted.

— END OF MODULE 8 SPECIFICATION —

MODULE 9 — METEOROLOGICAL MODULE

Meteorological Module

START: 06:00 Daily — Met Data Collection Initiated

▼

◆ **IMD Met Department Established at Branch?**

▼

SCENARIO A: AWS Data Streams into AMS Every 60 Seconds

METAR Auto-Generated Every 30 Min | TAF 4× Daily | SPECI on Change

▼

SCENARIO B (Current): Auto-Fetch Nearest METAR (VIKR / VICG)

+ Manual Local Observation by DCFI (Wind | Vis | Cloud Base)

▼

AMS Generates Formatted Weather Summary with Mandatory Disclaimer

▼

◆ **Weather Meets VFR Minima? (Vis ≥5km | Ceiling ≥1,500ft)**

▼

CFI Signs Off Go/No-Go → Flying Programme Confirmed or HOLD Declared

▼

END: Weather Record Archived — Immutable Audit Trail for Each Sortie

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MODULE 9 | METEOROLOGICAL MODULE

METEOROLOGICAL MODULE

9.1 Current Status and Regulatory Context

Neither HICA Karnal nor HICA Pinjore currently has a formal meteorological department. This represents a significant safety and compliance gap. DGCA regulations for FTOs require that adequate weather information be available to pilots for flight planning and pre-flight briefing. The AMS Met Module addresses this through two operational scenarios.

9.2 Scenario B — Interim Arrangement (Current Status)

- Auto-pull nearest METAR from AAI data feed: VIKR (Karnal), VOCX (Chandigarh) for Pinjore
- Displayed with mandatory disclaimer: 'Data from nearest aerodrome — conditions at HICA aerodrome may differ. Responsible officer must confirm local conditions before authorizing flight.'
- Manual local observation entry by DCFI/CFI: wind direction/speed (anemometer reading), visibility estimate, cloud base (if available)
- METAR-format display for pilot briefing with auto-archival for compliance records
- System tracks missing met data as a compliance gap and escalates to Safety Officer

9.3 Scenario A — Recommended: Establish IMD Met Department

- AWS (Automatic Weather Station) data ingestion in real-time (temp, dew point, pressure, wind, visibility, cloud)
- METAR generation every 30 minutes; SPECI on significant change; TAF 4 times daily
- Integration with pilot briefing workflow: Met Officer briefs CFI/DCFI; briefing log recorded with signature
- SIGMET/AIRMET relay from AAI/IMD higher offices

MODULE 10 — SAFETY MANAGEMENT SYSTEM

Safety Management System

START: Hazard / Incident Reported (Any Personnel, Any Device)

▼

Hazard Logged: Location | Phase | Description | Consequence | Action Taken

▼

Safety Officer Assesses Risk: 5×5 Matrix (Severity × Likelihood)

▼

◆ **Risk Level: RED (Unacceptable) / AMBER / GREEN?**

▼

RED: Mandatory AAIB + DGCA Notification | Investigation Team Assigned

Root Cause Analysis (ICAO Doc 9756) | CAPA Raised with Due Dates

▼

Safety Review Committee Quarterly Meeting Tracked

Minutes Recorded | Action Items Assigned

▼

ERP Maintained & Annual Exercise Conducted — Compliance Calendar Tracked

▼

END: Risk Register Updated — SMS Compliance Dashboard Refreshed

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MODULE 9 | SAFETY MANAGEMENT SYSTEM (SMS)

SAFETY MANAGEMENT SYSTEM (SMS)

9.1 Hazard Reporting

Any personnel — including students — can raise a confidential or named hazard report from any device via a single-tap button. Report fields capture: date and time, location, phase of operations, hazard description, potential consequence, immediate action taken, and optional reporter details for confidential submissions. This low-barrier reporting mechanism is fundamental to building a positive safety culture.

9.2 Risk Register

The Safety Officer owns the risk register, which uses a 5×5 risk matrix (Severity × Likelihood) to classify each hazard as Unacceptable (Red), Tolerable with Mitigation (Amber), or Acceptable (Green). Monthly reviews are mandatory. The register is fully searchable and supports filtering by risk level, phase of operations, and responsible officer.

9.3 Incident/Accident Investigation Workflow

- Any incident/accident triggers mandatory notification to: AAIB (per Aircraft Rules), DGCA, insurance
- Investigation team assigned; root cause analysis using ICAO Doc 9756 methodology
- Corrective and Preventive Actions (CAPA) tracked with owner, due date, and verification sign-off
- Investigation report template pre-built to DGCA/AAIB format

9.4 Safety Meeting Management and ERP

- Quarterly Safety Review Committee (SRC) meetings scheduled and tracked
- Minutes recorded in-system; action items assigned with due dates
- Annual Safety Day planning and execution tracked
- Full ERP stored in-system for each aerodrome
- Annual ERP exercise scheduling and compliance tracking
- Contact directory: fire brigade, ambulance, police, DGCA emergency, insurance, next-of-kin contacts

MODULE 11 — LEGAL & COMPLIANCE

Legal & Compliance

START: Legal / Regulatory Event Received (DGCA Notice | RTI | Court Case)

▼

Case Registered: Case No. | Authority | Responsible Officer | Status

▼

Automated Hearing Calendar: Reminders at T-14 | T-7 | T-1 Days

▼

◆ **RTI Request? (30-Day Statutory Response Deadline Tracked)**

▼

DGCA Show-Cause / Inspection Findings Tracked → Compliance Report Filed

▼

Legal Document Vault: Contracts | MOUs | Regulatory Approvals Stored

▼

◆ **Escalation Required? (Director Alert at T-3 Days)**

▼

END: Case Closed / Resolved — Record Archived in Legal Vault

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MODULE 10 | LEGAL & COMPLIANCE

LEGAL & COMPLIANCE

10.1 Case Registry

All legal cases, consumer complaints, and RTI requests are logged with a case number, court or authority, next hearing date, responsible officer, and current status. The Case Registry provides a complete legal history of the organisation, ensuring no hearing or deadline is missed.

10.2 Hearing Calendar and RTI Management

Automated hearing reminders are sent 14 days, 7 days, and 1 day before any hearing or regulatory deadline, with escalation to the Director at each stage. RTI requests are tracked to the 30-day statutory response deadline with auto-escalation to the Director if response is not logged in the system within the deadline.

10.3 DGCA Correspondence Tracker

All DGCA show-cause notices, inspection findings, compliance reports, and correspondence are tracked in a dedicated sub-module. Each item has a response due date, responsible officer, and evidence attachment capability. The tracker ensures that no DGCA communication goes unanswered and that all responses are documented with a complete audit trail.

10.4 Legal Document Vault

All contracts, agreements, MOUs, regulatory approvals, and legal documents are stored in a version-controlled Legal Document Vault with access restricted by role. Document version history allows comparison between versions, and automated expiry alerts are triggered for any document with a defined validity period.

MODULE 12 — HOSTEL MANAGEMENT

Hostel Management

START: Student Enrolment Triggers Hostel Room Allocation Request

▼

Student Coordinator Assigns Bed-Level Room (Occupancy Dashboard Live)

▼

Check-In Recorded | Visitor Log Active | CCTV Integration Enabled

▼

Daily Operations: Mess Menu | Meal Headcount | Maintenance Requests via App

▼

Security Rounds: QR Code Sign-Off at Checkpoints | Incident Reporting

▼

◆ **Curfew Breach Detected?**

▼

Auto-Alert: Warden + Student + Parent/Guardian Notified via SMS

▼

Exception Logged: Warden Approval Required for Authorised Late Returns

▼

END: Hostel Records Updated — Monthly Occupancy & Cost Report Generated

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MODULE 12 | HOSTEL MANAGEMENT

HOSTEL MANAGEMENT

12.1 Room Allocation and Occupancy

- Room Allocation: Bed-level assignment by Student Coordinator with check-in/check-out date tracking
- Occupancy Dashboard: Real-time room occupancy, vacancy, and maintenance status

12.2 Visitor Log and Security

- All visitor entries are recorded with date, time, purpose, and sign-out, with optional CCTV integration for entry-point cameras.
- Night security rounds are logged via QR code sign-offs at defined checkpoints, creating a verifiable security audit trail.
- Any security incident is reported through the hostel module and escalated to the Safety Officer.

12.3 Mess Management and Maintenance

- Weekly menu planning, vendor management, daily meal headcount, and cost tracking are managed within the Mess Management sub-module.
- Students can raise maintenance requests via the app, and the Hostel Warden tracks all requests from submission to resolution.
- Curfew times are configurable with exceptions requiring Warden approval — any breach triggers an automated alert to the Warden, student, and Parent/Guardian simultaneously.

MODULE 13 — FINANCE & ACCOUNTING

Finance & Accounting

START: Financial Transaction Initiated (Fee | Vendor | Payroll | Petty Cash)

▼

REVENUE: Student Fee Collected via Cash / NEFT / UPI

GST E-Invoice Auto-Generated | Fee Ledger Updated

▼

EXPENDITURE: PO → Vendor Payment Workflow

Payroll: TDS | PF | ESI | Payslip Generation

▼

Daily Bank Reconciliation via API | Petty Cash Imprest Reconciliation

▼

◆ **Student Fee Overdue? Outstanding Balance Alert?**

▼

Auto-Reminder at T-7 | T-3 | T-1 Days → Defaulter Flag → CFI Override Required

▼

Per-Branch P&L Generated | GST Returns Filed | Statutory Audit Package

▼

END: Finance Module Updated — Director Dashboard Refreshed

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MODULE 13 | FINANCE & ACCOUNTING

FINANCE & ACCOUNTING

13.1 Revenue Management

- Course fee collection: cash, NEFT, UPI, cheque; instant digital receipt (GST invoice)
- Fee structure management: configure different fee slabs (theory only, full CPL, etc.)
- Outstanding fee alerts: automated reminders at 7, 3, 1 day before due date

13.2 Expenditure Management

- Vendor payment management: PO to payment workflow; MSME payment compliance (45-day rule)
- Payroll: Salary register, TDS calculation, payslip generation, PF/ESI integration
- Petty cash: Imprest account management with daily reconciliation

13.3 GST, Bank Reconciliation, and Reporting

- GST: E-invoice generation, GSTR-1/3B auto-calculation, HSN/SAC code management
- Bank Reconciliation: Daily automated reconciliation via bank statement API
- Per-Branch P&L: Separate profit & loss accounts for Karnal and Pinjore with consolidation at HICA level
- Budget vs. Actuals: Monthly variance reports for each cost centre
- Annual Statutory Audit package: all reports in CA-ready format

MODULE 14 — HUMAN RESOURCES

Human Resources

START: New Employee Joining Triggers HR Onboarding Workflow

▼

Employee Master Created | Documents Uploaded | MFA Login Issued

▼

Role-Specific Onboarding Checklist Activated (FI / LAME / Safety Officer)

▼

Daily Attendance: Biometric / Geo-Fenced App Check-In

Leave Management: EL | CL | ML | Comp-Off Tracked

▼

Annual Appraisal: KPI-Based | Linked to Salary Revision Workflow

▼

◆ **Any Licence / Certificate Expiring? (Tracked Across All Staff)**

▼

Automated Reminder to Holder + CFI/Director at 90/60/30 Days

▼

Offboarding: Asset Recovery | Access Revocation | Exit Checklist Closed

▼

END: HR Record Updated — Compliance Calendar Populated

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MODULE 14 | HUMAN RESOURCES

HUMAN RESOURCES

14.1 Employee Master and Onboarding

- The Employee Master contains a complete profile per the Part 5 personnel matrix — qualifications, licences, certificates, and contact details.
- When a new employee is created, the system triggers a structured onboarding workflow: welcome email with temporary credentials, mandatory MFA setup, and a role-specific onboarding checklist (for example, a Flight Instructor must upload CPL/ATPL licence, FI Rating, and Class 1 Medical scans, and complete the SMS awareness briefing before full system access is granted).

14.2 Attendance and Leave Management

- Attendance is captured via biometric integration (Aadhaar-based UIDAI or standalone device) or geo-fenced mobile app check-in for field personnel.
- The leave management system handles Earned Leave, Casual Leave, Medical Leave, and Compensatory Off with automated balance calculation and an approval workflow routed to the reporting manager.
- Leave approvals or rejections trigger automated notifications to the employee.

14.3 Appraisal, Recruitment, and Offboarding

- The annual and half-yearly KPI-based appraisal system sends automated reminders to reporting managers, stores ratings, and links to the salary revision workflow.
- Recruitment management covers vacancy posting (with job portal integrations), applicant tracking, interview scheduling, and offer letter generation.
- Offboarding checklists ensure systematic asset recovery, IT access revocation, and document handover. All employee licences and certificates are tracked with automated reminders matching the DGCA Compliance Calendar cadence.

MODULE 15 — DGCA COMPLIANCE CALENDAR

DGCA Compliance Calendar

START: All 35 DGCA Compliance Obligations Loaded into Calendar

▼

Compliance Health Dashboard: RED | AMBER | GREEN Status per Obligation

▼

Automated Alerts at T-90 | T-60 | T-30 | T-14 | T-7 | T-1 Days

▼

◆ **T-7 Days: Item Not Marked Complete? Escalate to CFI**

▼

◆ **T-3 Days: Still Incomplete? Escalate to Director**

▼

Responsible Officer Completes Action → Evidence Attached (e.g., C of A Scan)

▼

DGCA Audit Readiness Report Generated On-Demand: One-Page Compliance Status

▼

END: Compliance Record Updated — Dashboard Status Refreshed

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MODULE 15 | DGCA COMPLIANCE CALENDAR

DGCA COMPLIANCE CALENDAR

Full compliance calendar provided in Section 11. Module functions:

- All recurring DGCA obligations tracked with due dates, responsible person, and documentation requirements
- Automated alerts at 90, 60, 30, 14, 7, and 1 day(s) before due date
- Escalation matrix: if item not marked complete at T-7 days, alert goes to CFI; at T-3 days, to Director
- Compliance health dashboard: Red/Amber/Green status for every tracked obligation
- Evidence vault: Attach compliance proof (e.g., scan of renewed C of A, completed inspection form)
- DGCA audit readiness report: generates a one-page compliance status report instantly on demand

MODULE 16 — FLIGHT PLANNING

Flight Planning

START: Instructor / DCFI Initiates Flight Plan for Upcoming Sortie

▼

Route Entry on Built-in VFR Planner or Import from Garmin Pilot (GPX/FPL)

▼

Performance Calculator: T/O Distance | Climb | Cruise TAS | Fuel Burn | Endurance

(Using C152 / C172 POH Data from Fleet Module)

▼

NOTAM Overlay: Auto-Fetch for Departure | Route | Destination Aerodromes

▼

Fuel Plan: Trip + Alternate + Reserve (per DGCA CAR) → Uplift Requirement

▼

W&B Calculation: Gross Weight + CG Plot → GO / NO-GO (per Module 1)

▼

◆ **ICAO FPL Approved by CFI — All Checks Passed?**

▼

END: Flight Plan Filed | Linked to Sortie Record for Post-Flight Reconciliation

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MODULE 16 | FLIGHT PLANNING

FLIGHT PLANNING

16.1 Garmin Pilot Integration and Built-in VFR Planner

- Garmin Pilot Integration: Import flight plans from Garmin Pilot via shared plan URL or GPX/FPL export
- Built-in VFR Planner: Route entry on map, automatic ICAO FPL generation (Format per ICAO Doc 4444)
- W&B: Integrated with Module 1 Go/No-Go (see 1.3)

16.2 Performance Calculator

- Performance Calculator: T/O distance, climb gradient, cruise TAS, fuel burn, endurance — using POH data for C152/C172 stored in Fleet module

16.3 NOTAM Overlay, Weather, and Fuel Planning

- NOTAM Overlay: Auto-fetch and display NOTAMs for departure, route, destination aerodromes
- Weather Overlay: Route-segment weather briefing from MET module or external API
- Fuel Planning: Calculate required fuel (trip + alternate + reserve per DGCA CAR) and uplift requirement
- Alternate Aerodrome Selection: Nearest aerodromes displayed with METAR; CFI selects appropriate alternate

MODULE 17 — DIGITAL LOGBOOK

Digital Logbook

START: Sortie Completed — Post-Flight Data Submitted by Instructor

▼

Logbook Entry Auto-Populated: T/O | Landing | Aircraft | Exercise | Dual/Solo

(DGCA Schedule III Format — Indian Aircraft Rules 1937)

▼

Instructor Digital Sign-Off via OTP or Biometric

(Legally Valid under IT Act 2000)

▼

Entry Locked — Tamper-Proof: Corrections Require CFI-Authorized Amendment

▼

Career Summary Auto-Updated: TTAF | Dual | Solo | XC | Night | Instrument | PIC

▼

◆ **Student Requesting Transfer to Another FTO?**

▼

Digitally Signed JSON Export Package Generated for Destination FTO Import

▼

END: Logbook PDF Export Available for DGCA Submission at Any Time

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MODULE 17 | DIGITAL LOGBOOK

DIGITAL LOGBOOK

17.1 DGCA Format and Auto-Population

- The Digital Logbook is maintained in strict compliance with the format prescribed in Indian Aircraft Rules 1937 (Schedule III).
- All sortie data from Module 1 — take-off time, landing time, aircraft registration, exercise number, and dual/solo classification — automatically populates the logbook entry after the instructor's digital sign-off.
- This eliminates manual transcription errors and ensures every logbook entry is backed by verified system data.

17.2 Digital Sign-Off and Tamper-Proof Records

- Instructor digital sign-off is implemented via OTP or biometric confirmation, which is legally equivalent to a physical signature under the Information Technology Act 2000.
- Once signed, entries are locked — any correction requires a separate amendment record with a documented reason, authorised by the CFI — with the full audit trail preserving both the original and amended values.
- The immutable nature of the logbook provides the same evidentiary value as a physical logbook in any regulatory or legal proceeding.

17.3 Career Summary, PDF Export, and Portability

- Cumulative totals are auto-calculated and always current: TTAF, Dual, Solo, Cross-Country, Night, Instrument, and PIC hours.
- The student can generate a DGCA-format logbook PDF printout at any time for direct submission to DGCA.
- When a student transfers to another FTO, their complete logbook record can be exported as a digitally signed JSON package for import by the destination FTO if it is also on the AMS platform, enabling seamless portability without paper logbook risk.

MODULE 18 — COMMUNICATION ENGINE

Communication Engine

START: Trigger Event Detected (Schedule Change | Alert | Roster Update | Emergency)

▼

Communication Engine Identifies Relevant Recipients by Role & Branch

▼

Message Template Selected: Pre-Approved, Role-Specific, Multi-Language Ready

▼

◆ **CFI Acknowledgement Required Before Sending?**

▼

Multi-Channel Dispatch: Push Notification | SMS (MSG91) | WhatsApp Business | Email

▼

Delivery Confirmation Logged | Undelivered Messages Escalated

▼

Internal Messaging: Thread-Based per Aircraft / Sortie / Student Record

Broadcast Notice Board: CFI/Director Organisation-Wide Notices

▼

END: All Communications Logged with Timestamp — Full Audit Trail

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MODULE 18 | COMMUNICATION ENGINE

COMMUNICATION ENGINE

18.1 Automated Notifications — Multi-Channel

Trigger Event	Recipient(s)	Channel(s)
Tomorrow's flying schedule published	All assigned students & instructors	Push, WhatsApp, SMS
Aircraft U/S (snag raised)	CFI, DCFI, AME	Push, Email, SMS
Weather below VFR minima	CFI, all scheduled students/ instructors	Push, SMS
Student medical expires in 30 days	Student, CFI, Admin	Email, Push
Any license/cert expiring in 60 days	Holder, CFI/Director	Email, Push
Overdue aircraft alert	Dispatcher, CFI, DCFI, Emergency Contacts	SMS, Push, Email
Fee payment due in 7 days	Student, Parent/Guardian, Accounts	SMS, WhatsApp
Hostel curfew breach	Warden, Student, Parent/ Guardian	SMS, Push
New AD issued for aircraft	CFI, AME, Safety Officer	Email, Push
Fuel stock below reorder level	CFI, Storekeeper, Procurement	Push, Email

18.2 Communication Approval Protocol

- All automated messages generated by the AIRE are pre-approved templates. The CFI approves the communication cascade with a single tap after reviewing the revised roster — no message is sent without CFI acknowledgment.
- This ensures that the CFI remains the single point of authority for all operational communications while eliminating the manual effort of placing individual calls.

18.3 Internal Messaging

- Secure in-app messaging between all staff, respecting role hierarchy
- Thread-based discussions per aircraft, sortie, or student record
- Broadcast notice board: CFI/Director can publish organization-wide notices

MODULE 19 — REPORTING & ANALYTICS

Reporting & Analytics

START: Data Continuously Aggregated from All 19 AMS Modules

▼

Role-Based Dashboards: CFI | AME | Finance | Director

Real-Time KPIs: Sorties | Serviceability | Compliance | Revenue

▼

Standard Reports Auto-Generated:

Monthly (Utilisation | Fuel | Student Progress | Duty Time)

Quarterly (Safety | Finance P&L) | Annual (DGCA Returns)

▼

AI-Powered Insights:

Student CPL Trajectory Prediction | Maintenance Cost Forecast

Optimal Fleet Utilisation Recommendations

▼

◆ **AI Flags At-Risk Student (Below Hour Target Pace)?**

▼

CFI Alert Raised → AI Rostering Engine Adjusts Priority Score for Student

▼

DGCA Audit Package Generated On-Demand: One-Click Compliance Summary

▼

END: All Reports Stored with Immutable Timestamp — Available for Audit

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MODULE 19 | REPORTING & ANALYTICS

REPORTING & ANALYTICS

19.1 Operational Dashboards

- Role-based dashboards present the most operationally relevant information to each user: the CFI Dashboard shows today's sorties, aircraft availability, weather status, overdue alerts, and student hours summary; the AME Dashboard shows open snags, due maintenance, AD compliance status, and low-stock parts alerts; the Finance Dashboard shows fee collection month-to-date, payables outstanding, fuel expenditure, and branch P&L summary; and the Director Dashboard consolidates all key metrics — utilisation, compliance health, safety events, and financial summary — across both branches.

19.2 Standard Reports — Automated

- Monthly reports are generated automatically: aircraft utilisation, fuel consumption, student training progress, and instructor duty time.
- Quarterly reports cover safety review, ground training progress, and finance P&L. Annual reports include the DGCA Annual Return data pack (ready for direct submission), fleet maintenance cost, and staff performance summary. All reports are stored with immutable timestamps in the document vault.

19.3 AI-Powered Insights

- The AI insights engine provides three categories of predictive intelligence.
- First, student trajectory prediction flags students at risk of falling behind their CPL hour target based on current completion rate and historical pace — enabling proactive intervention before the gap becomes unrecoverable.
- Second, aircraft maintenance cost prediction uses a machine learning model to forecast the next six months of maintenance cost per aircraft, enabling accurate budget planning.
  Third, optimal fleet utilisation recommendations suggest the best windows to take each aircraft for scheduled maintenance, minimising operational impact by analysing historical flying patterns and upcoming student demand.

3. PERSONNEL MATRIX — CURRENT STATE & GAP ANALYSIS

3.1 HICA Personnel Matrix

The following table provides the complete personnel structure for HICA, including minimum DGCA requirements, current strength, and identified gaps. Gaps are classified by urgency.

Role	DGCA Mandatory?	Min Qualifications	Karnal (Cur/Req)	Pinjore (Cur/Req)	Gap Status
Chief Flying Instructor (CFI)	YES	CPL/ATPL + FI Rating + 200h dual instruction	1/1 ✓	0/1 ✗	CRITICAL — Pinjore
Deputy CFI (DCFI)	YES (if >12 students)	CPL/ATPL + FI Rating + 150h dual	1/1 ✓	1/1 ✓	Met
Flight Instructor (FI)	YES (min 1 per aircraft)	CPL + FI Rating	0/2 ✗	0/2 ✗	HIGH — Both
Check Pilot	Recommended	CPL/ATPL + FI + Check Pilot auth.	0/1 ✗	0/1 ✗	MEDIUM — Both
Chief Ground Instructor (CGI)	YES	CPL/ATPL + 3yr instructing exp.	0/1 ✗	1/1 ✓	HIGH — Karnal
Ground Instructor — Air Regs	YES	CPL + subject knowledge	0/1 ✗	1/1 ✓	HIGH — Karnal
Ground Instructor — Nav	YES	CPL + Navigation specialty	0/1 ✗	0/1 ✗	HIGH — Both
Ground Instructor — Met	YES	Met diploma or equivalent	0/1 ✗	0/1 ✗	HIGH — Both
Ground Instructor — Tech General	YES	AME or CPL + tech background	0/1 ✗	0/1 ✗	HIGH — Both
Ground Instructor — Tech Specific	YES	Cessna endorsement	0/1 ✗	0/1 ✗	HIGH — Both
Ground Instructor — RTF	YES	RTF Examiner authorization	0/1 ✗	0/1 ✗	HIGH — Both
LAME Cat B1 (A&P)	YES — CRITICAL	DGCA LAME License B1 (Aircraft Rules 1937)	0/1 ✗	0/1 ✗	CRITICAL — Both
LAME Cat B2 (Avionics)	Recommended	DGCA LAME License B2	0/1 ✗	0/1 ✗	MEDIUM — Both
Asst. AME / Mechanic	YES (under LAME)	Aircraft Maintenance Engineer (basic)	0/2 ✗	0/2 ✗	HIGH — Both
Aircraft Marshaller	Recommended	Trained marshalling personnel	0/1 ✗	0/1 ✗	LOW — Both
Met Officer	Recommended (Required if IMD)	Met degree + IMD authorization	0/1 ✗	0/1 ✗	HIGH — Both
Met Observer	If IMD established	IMD Observer training	0/1 ✗	0/1 ✗	MEDIUM — Both

Role	DGCA Mandatory?	Min Qualifications	Karnal (Cur/Req)	Pinjore (Cur/Req)	Gap Status
Safety Officer (SMS Focal)	YES — DGCA CAP 8100	Aviation background + SMS training	0/1 ✗	0/1 ✗	HIGH — Both
Director / Head of FTO	YES	Experience in aviation management	1/1 ✓	1/1 ✓	Met (shared)
Administrative Officer	No	Graduate + admin experience	1/1 ✓	0/1 ✗	MEDIUM — Pinjore
Accounts Officer	No	B.Com/CA inter	1/1 ✓	0/1 ✗	MEDIUM — Pinjore
IT Admin	No	B.Tech/Diploma IT	0/1 ✗	0/1 ✗	MEDIUM — Both
Storekeeper	No	Inventory management training	0/1 ✗	0/1 ✗	MEDIUM — Both
Student Coordinator	No	Graduate + communication skills	1/1 ✓	1/1 ✓	Met
Hostel Warden	No	Graduate + residential management	1/1 ✓	1/1 ✓	Met

3.2 Gap Summary

Severity	Branch(es)	Roles	Action Required
CRITICAL	Pinjore	CFI	Hire immediately; Pinjore cannot independently operate as FTO without CFI
CRITICAL	Both	LAME Cat B1	Hire immediately; no aircraft can legally be certified for flight without LAME sign-off (CRS)
HIGH	Both	Flight Instructors (FI)	Minimum 2 per branch needed for efficient operations; 1 per aircraft is minimum DGCA
HIGH	Karnal	CGI + Ground Instructors	Ground training cannot be delivered without qualified GIs; theory exam eligibility at risk
HIGH	Both	Safety Officer	DGCA CAP 8100 requires a designated SMS Focal Point; non-compliance risks FTO renewal
HIGH	Both	Met Officer / Observer	Weather information quality is a safety-critical gap; pilots relying on remote METAR only

4. AIRCRAFT MANAGEMENT — FULL SPECIFICATION

4.1 Aircraft Identity & Documentation

Data Field	Description / Validation Rules
Registration Mark	VT-XXX format; validated against DGCA aircraft register API
Aircraft Manufacturer & Model	Cessna 152 / Cessna 172 (series); dropdown with model variant (172S, 172R, etc.)
Serial Number (Airframe)	Factory S/N; used for AD compliance matching
Year of Manufacture	4-digit year; used for AD age determination
Certificate of Airworthiness (C of A)	Issue date, expiry date, C of A number, category; auto-alert 90/60/30 days prior
Certificate of Registration (C of R)	Issue date, expiry date, issuing authority (DGCA Regional Office)
Certificate of Release to Service (CRS)	Issue date, CRS number, LAME who signed, hours at issue; required before every flight
Insurance Policy	Policy number, insurer, hull value, third-party liability cover amount, expiry date
AFM/POH	Document reference, revision number, date; PDF stored in document vault
MEL (Minimum Equipment List)	DGCA-approved MEL document; revision tracked; MEL items can be raised against open snags
Journey Log (Form CA 48)	Digital equivalent; all sortie entries from Flight Ops Module auto-populate
Technical Log	Maintenance entries; CRS records; AME sign-off; auto-populated from Maintenance Module

4.2 Airframe Tracking

Parameter	Tracking Method
TTAF (Total Time in Service)	Auto-updated from sortie records; manual override with AME authorization
Hours Since Last 50-Hour Check	Calculated from maintenance records; due at TTAF + 50 (± 5 hours tolerance)
Hours Since Last 100-Hour Check	Calculated; due at TTAF + 100
Hours Since Last Annual Inspection	Calendar-based (12 months from last Annual); and hour-based (whichever is first)
Next 50-Hour Due Date	Displayed on aircraft summary card with hours remaining
Next 100-Hour Due Date	Displayed; auto-triggers work order in Maintenance Module when <10 hours remaining
Next Annual Due Date	Calendar display; triggers 30-day advance alert to CFI and AME

Parameter	Tracking Method
Major Repair History	Free-text log per repair event; supports attachment of Repair Major Order (RMO) documents
Airframe Life Limit (if applicable)	Cessna single-engine: no hard life limit; but structural inspection tracking

4.3 Engine Tracking

Parameter	C152 (Lycoming O-235)	C172 (Lycoming O-320/O-360)
Manufacturer	Lycoming	Lycoming
Model	O-235-L2C	O-320-H2AD / O-360-A4M (per variant)
TBO (Time Between Overhaul)	2,400 hours (Lycoming)	2,000 hours (Lycoming)
Serial Number	Tracked; AD compliance matched	Tracked; AD compliance matched
TSN (Time Since New)	Cumulative hours since manufacture	Cumulative hours since manufacture
TSOH (Time Since Overhaul)	Hours since last overhaul; due before TBO	Hours since last overhaul
Last Oil Change	Date + TTAF at change; oil type (AeroShell W80)	Date + TTAF; oil type (AeroShell W100 Plus)
Oil Consumption Log	Quarts added per refill; trend analysis	Quarts added per refill; trend analysis
Compression Test Results	All 4 cylinder readings (C152); pass >60/80	All 4/6 cyl readings; pass >60/80
Mag Timing Records	Date, TTAF, timing setting, technician	Date, TTAF, timing setting, technician

4.4 Propeller Tracking

Parameter	Details
Manufacturer	Sensenich (C152: 69/63; C172: 76/58 or McCauley 1C160)
Serial Number	Tracked per aircraft; AD matching (especially for propeller strike ADs)
TSN	Total time since new (hours)
TSOH	Time since overhaul; overhaul due per manufacturer calendar interval (e.g., 72 months or 2,400 hrs)
TBO	Per manufacturer service letter/document
Strike History	Any ground strike / over-speed event logged with date, description, inspection result; prop strike requires IMMEDIATE inspection of prop and engine
Nick/Damage Records	Dress-out records: nick location, depth before/after dressing
Next Inspection Due	Calendar-based + hours-based; whichever is first

4.5 Avionics & Instruments Register

Each instrument and avionics unit is tracked individually in the Aircraft Instruments sub-module:

Instrument/Equipment	Make/Model (Typical)	Tracked Parameters
Airspeed Indicator (ASI)	Sigma Tek / United Instruments	S/N, last pitot-static test, next due
Attitude Indicator (AI)	Sigma Tek vacuum-driven	S/N, overhaul due, last inspection
Altimeter	Sigma Tek / United — 3-pointer	S/N, last pitot-static + encoding test, next due (24 months)
Vertical Speed Indicator (VSI)	United Instruments	S/N, last pitot-static test
Directional Gyro (DG)	Sigma Tek vacuum-driven	S/N, overhaul due, precession check log
Magnetic Compass	Airpath C-2300 or similar	S/N, last swing/deviation card update, mount security
Tachometer	Cessna electric or mechanical	S/N, calibration date
Oil Pressure Gauge	Cessna standard	S/N, last calibration
Oil Temperature Gauge	Cessna standard	S/N, last calibration
EGT / CHT	Alcor / Insight or Cessna OEM	S/N, last calibration, probe condition
Fuel Quantity Gauges (L+R)	Cessna resistive gauges	S/N, last calibration vs. tank dip check
Ammeter/Loadmeter	Cessna standard	S/N, last inspection
Suction Gauge (Vacuum)	Cessna standard	S/N, acceptable range (4.5–5.4 inHg)
VHF Comm Radio #1	Garmin GMA 340 or GTR 225	S/N, last check, frequencies programmed
VHF Comm Radio #2	Garmin GTR 225 or King KX155	S/N, last check
VOR/ILS Nav #1	Garmin GNC 300 or KX165	S/N, last VOR check (per FAR 91.171), accuracy
GPS/Moving Map	Garmin GTN 650 / GNS 430W or GNS 530W	S/N, database subscription expiry, RAIM availability
Transponder	Garmin GTX 327 or GTX 335 (ADS-B)	S/N, last biennial transponder check (24 months), Mode C altimetry check
ELT	ACK E-04 or Artex ME406 (406 MHz)	S/N, battery expiry, antenna condition, last annual inspection
Marker Beacon Receiver	If fitted	S/N, last check
Audio Panel / Intercom	Garmin GMA 340	S/N, last inspection

Instrument/Equipment	Make/Model (Typical)	Tracked Parameters
Pitot Heat System	Cessna OEM	Functional test log, element resistance check

4.6 Safety Equipment Tracking

Equipment	Fields Tracked
Fire Extinguisher	Make, S/N, type (Halon 1211 or CO2), capacity, last hydro test, next hydro test, last inspection, charge status
First Aid Kit	Contents checklist, last inspection, expiry dates of medications, replenishment log
Seat Belt — Pilot Seat	Make, date of manufacture, last inspection, condition (per FAA AC 91-9), next replacement due
Seat Belt — Co-Pilot Seat	Same as above
Seat Belt — Rear Seat (L)	Same as above
Seat Belt — Rear Seat (R) [C172 only]	Same as above
Shoulder Harness	Inspection status, per-seat tracking
Emergency Locator Transmitter (ELT)	Tracked in Avionics section; also appears in Safety Equipment for battery management

4.7 Airworthiness Directive (AD) Compliance

4.7.1 AD Database

- Airframe ADs: Queried against FAA AD database (rgl.faa.gov) using aircraft manufacturer + model + S/N
- Engine ADs: Queried using engine manufacturer + model + S/N (e.g., Lycoming SSP-110, SSP-113)
- Propeller ADs: Queried using prop manufacturer + model + S/N
- Avionics ADs: Where applicable (e.g., Garmin GTN AD for software vulnerability)
- DGCA ADs: Separate queue from DGCA website; DGCA-issued ADs override FAA ADs where applicable

4.7.2 AD Tracking Fields

Field	Description
AD Number	FAA or DGCA AD reference number
Title	Short description
Effective Date	Date AD became effective

Field	Description
Applicability	Confirms aircraft S/N falls within applicability (system checks automatically)
Compliance Time	One-time, recurring, on-condition, terminating
Due At (TTAF)	Hours at which next compliance is due
Compliance Method	Per the AD (alteration, inspection, replacement — with AMM reference)
Last Complied At (TTAF)	Time at last compliance
Next Compliance Due	Calculated automatically
Sign-Off	LAME who performed compliance, date, CRS reference
Status	Open / Complied / N/A (with reason)

4.8 Scheduled Maintenance — Checklist Framework

The AMS contains pre-built maintenance checklists for Cessna 152 and 172 aligned with the Cessna Aircraft Maintenance Manual (AMM). Key checks:

Check Type	Interval	Key Tasks (Summary)
50-Hour Inspection	Every 50 flight hours	Oil change, spark plug rotation, air filter check, control system inspection, operational check of all instruments, landing gear inspection, magneto timing check
100-Hour Inspection	Every 100 flight hours (also serves as Annual for hire aircraft)	All 50-hour tasks PLUS: complete airframe inspection, engine compartment detailed inspection, compression test on all cylinders, pitot-static functional check, avionics check, brake pad measurement, tyre check, battery check
Annual Inspection	Every 12 calendar months	Equivalent to 100-hour; mandatory regardless of hours flown; must be signed off by LAME; includes airworthiness review for C of A renewal
Engine Overhaul	At TBO (2,400/2,000 hrs) or as required	Complete engine tear-down, inspection of all components to new limits, replacement of all life-limited parts, test run, re-installation
Propeller Overhaul	Per manufacturer interval or at damage	Disassembly, dimensional inspection, repair as required, dynamic balance, reinstallation
Avionics Biennial	Every 24 months	Transponder altimetry check, VOR accuracy check, pitot-static system leak test, ELT inspection

4.9 Snag Management Workflow

The Snag Management workflow is a complete digital replacement for the paper-based Snag Book:

1. SNAG RAISE: Pilot (post-flight) or AME (during inspection) creates a Snag Record. Fields: Aircraft reg, date/time, sortie (if in-flight), description of defect, affected system, MEL reference (if applicable), severity (AOG / Schedule / Deferred)
2. INITIAL ASSESSMENT: CFI reviews. AOG snags immediately ground the aircraft and remove it from dispatch eligibility. Scheduled snags get a target rectification date.
3. AME ASSIGNMENT: CFI/DCFI assigns the snag to the LAME with target completion date. AME acknowledges via app.
4. RECTIFICATION: AME opens a Work Order linked to the snag. Enters: AMM reference, part numbers used (linked to Parts Inventory for stock deduction), man-hours. Attaches photos of defect and repair.
5. QUALITY CHECK: For complex snags, a second AME (or LAME B2 for avionics) performs a quality inspection and signs off.
6. CRS ISSUANCE: LAME issues Certificate of Release to Service via the AMS. CRS is digitally signed, assigned a unique reference number, and linked to the snag record, the work order, and the aircraft's technical log. Aircraft returned to service in AMS.
7. CLOSURE: Snag record closed. Retained in aircraft history. If recurring, Safety Officer automatically notified for trend analysis.

4.10 Parts & Consumables Inventory

Category	Items Tracked
Lubricants	AeroShell W80 (C152 engine), AeroShell W100 Plus (C172 engine), AeroShell Grease No.6 (airframe), AeroShell Fluid 41 (hydraulic)
Filters	Engine oil filter (C152: Cessna 0756-5501, C172: Cessna 0756-5501-06), Fuel strainer screens
Spark Plugs	Champion REM37BY or Tempest UREM37BY (C152), Champion RHB32E or Tempest URHB32E (C172); track by cylinder position
Tyres	Main: 6.00-6 (C152), 6.00-6 (C172); Nose: 5.00-5; stock qty, expiry (DOM + shelf life)
Brake Pads	Cleveland brake pad P/N per aircraft; minimum thickness threshold alert
Batteries	Concorde RG-35AXC (C152) or RG-25XC (C172); installation date, equalization charge log, load test log
ELT Batteries	ACK or Artex replacement battery; expiry date per installed ELT batch code
Bulbs & Fuses	Nav light bulbs (GE1683, GE4557), strobe power supply fuses, circuit breakers inventory
Safety Wire	0.020", 0.032", 0.041" stainless; qty in meters

Category	Items Tracked
Hardware	AN bolts, AN nuts, cotter pins (AN380, AN381 series); stock levels
Seals & Gaskets	Engine oil seals, fuel cap gaskets; stock by P/N
Fuel Caps & Gaskets	Cessna fuel cap gaskets; inspection and replacement at 100-hour/ Annual

Each part has: Part Number, Description, Unit, Reorder Point (minimum stock), Reorder Quantity, Current Stock, Last Purchase Price, Preferred Vendor. When stock falls to or below the Reorder Point, the system auto-generates a Purchase Requisition to the Procurement Officer.

5. FUEL MANAGEMENT — FULL SPECIFICATION

5.1 Fuel Type Enforcement

The AMS enforces AVGAS 100LL as the sole authorized fuel for all HICA aircraft (Lycoming O-235, O-320, O-360 engines all require 100LL minimum, minimum octane 100 for detonation margin at high power settings). Any attempt to log or procure Jet-A, AVTUR, or MOGAS triggers a system-level CRITICAL ERROR with visual and audio alarm, and blocks the action until an authorized officer explicitly acknowledges and overrides with a documented reason. MOGAS use would require an STC (Supplemental Type Certificate) which HICA aircraft do not hold.

5.2 Procurement Workflow

Step	Trigger	Actions	System Record
Auto-PR Generation	Stock < Reorder Level	System creates draft Purchase Requisition; alerts Procurement Officer and CFI	PR Number, timestamp, current stock, reorder level
PR Approval	Procurement Officer action	CFI/Director approves PR with quantity and supplier selection	Approval timestamp, approver, authorized quantity
PO Issuance	PR approved	System generates formal PO to fuel supplier (IOC Aviation/ BPCL as applicable)	PO Number, supplier, quantity, unit price, delivery date
Delivery Receipt	Fuel tanker arrives	Storekeeper/AME logs: delivery date, time, supplier name, vehicle reg, driver name, delivery note number, batch/lot number, quantity delivered (liters), temperature at delivery, observed density (kg/L)	
Quality Acceptance Test	Before defueling to storage	Mandatory 5 tests before acceptance (see 6.3)	Test results per batch
Storage Entry	QAT passed	Quantity added to specific storage (fixed tank/bowser/ drum); running stock updated	Stock record, batch traceability
Rejection & Quarantine	QAT failed	Fuel quarantined; supplier notified; DGCA notification if contamination suspected; disposal record required	Quarantine log, disposal certificate

5.3 Quality Acceptance Testing

Test	Method	Pass Criteria	Failure Action
Visual — Color	Sample in clear glass container vs. daylight	Clear blue color (100LL is dyed blue); no cloudiness	Quarantine; re-test; reject if repeated
Visual — Clarity	Same sample; inspect for solids, particulate	Bright and clear; no visible contamination	Quarantine; filter test
Water Detection	Millipore water detector or AquaGlo	No free water; <15 ppm dissolved (AquaGlo limit)	Drain sump; re-test; reject if persistent
Density Check	Calibrated hydrometer in temperature-corrected sample	0.695–0.721 kg/L at 15°C (per ASTM D6615)	If out of range: quarantine; laboratory analysis
Filter Condition	Inspect inline filter element after delivery	No color change (water-absorbing element stays white/ beige)	Replace filter; investigate delivery vehicle

All test results are logged with: tester's name, date/time, batch number, instrument calibration reference. The system will not move fuel from 'Delivered — Pending QAT' status to 'Available Stock' until all 5 tests are recorded as PASS.

5.4 Storage Tracking

Storage Type	Fields Tracked
Fixed Underground Tank	Capacity (liters), current stock (liters), gauge calibration date, tank inspection date, integrity test date, ground bonding continuity check date
Fuel Bowser / Tanker (mobile)	Registration/ID, capacity, current fuel level, last calibration of meter/flowmeter, bonding reel condition, filter element type and last change, next service due
Fuel Drums	Drum number, capacity, current quantity (estimated by dipstick), supplier, batch number, seal integrity, storage condition (upright, shaded, on bund)

Monthly Compliance Inspections

Fuel facility monthly inspection tracks:

- Bonding cable continuity (aircraft-to-bowser, bowser-to-ground)
- Fire extinguisher availability and status (minimum 2 × 9kg DCP at fueling point)

- Spill kit completeness (absorbent pads, containment berm)
- No smoking signs in place and visible
- Fuel storage area clear of ignition sources
- Bund wall integrity (for fixed tank installations)

Monthly inspection is a mandatory Compliance Calendar item assigned to the Safety Officer with CFI co-sign.

5.5 Pre-Flight Fuel Checks (Per Sortie)

Recorded by the pilot or line technician before each sortie:

- Sump drain check: both wing tank sumps and gascolator drained; visual check for water or contamination; color confirmed blue
- Fuel cap security: both caps latched and sealed
- Fuel quantity confirmed (dipstick or sight gauge) vs. required for sortie
- Uplift quantity recorded if fuel added before this check

5.6 Fuel Uplift Per Sortie

- Uplift quantity entered post-sortie: liters added, bowser meter start/end reading
- Consumption calculation: Opening quantity + Uplift − Closing quantity = Consumption
- Planned consumption: calculated from flight planning (POH fuel burn rate × planned duration)
- Variance alert: If actual consumption > planned by 10%: AMBER alert to CFI (possible fuel leak, engine running rich). If >20%: RED alert with mandatory investigation note.

5.7 Monthly Fuel Audit

Automated at month-end:

- Total uplifts received vs. supplier invoices
- Total sorties uplift vs. logbook fuel entries
- Stock reconciliation: opening + receipts − sorties usage = closing stock (vs. physical dip measurement)
- Loss/gain report: acceptable tolerance ±2% (accounting for temperature variation)
- If variance exceeds 2%: mandatory investigation assigned to Safety Officer

6. INSTRUMENTS & GROUND EQUIPMENT — FULL TRACKING DESIGN

6.1 Ground Equipment Asset Register

Every item of ground equipment is an asset record in the AMS with: Asset ID (barcode/QR code), description, category, location (branch + specific area), status (Serviceable / U/S / Calibration Due), purchase date, cost, vendor, warranty expiry, next inspection due, and custodian.

6.2 Aircraft Handling Equipment

Equipment	Per Branch Qty	Inspection Interval	Key Parameters
Tow Bar (C152)	1	Annual	Attachment pin condition, tow ring wear, no cracks
Tow Bar (C172)	1	Annual	Same as above; verify nose gear adapter fit
Wheel Chocks (pairs)	6	6 months	Rubber condition, no cracking, rope/strap serviceability
Tie-Down Kit (3-point)	6 sets	6 months	Rope/strap tensile test, hardware corrosion check
Pitot Tube Cover (per aircraft)	1 per aircraft	6 months	Flag condition, fit to pitot tube, 'REMOVE BEFORE FLIGHT' tag legible
Control Lock (yoke/rudder)	1 per aircraft	6 months	Lock mechanism serviceable, 'REMOVE BEFORE FLIGHT' tag
Intake Cover (engine cowl)	1 per aircraft	Annual	Fit and retention cord condition

6.3 Fueling Equipment

Equipment	Inspection Interval	Key Compliance Parameters
Fuel Bowser (mobile)	Monthly (operational) + Annual (statutory)	Meter calibration, filter change log, bonding reel, fire extinguisher mounted
Hand Pump / Transfer Pump	Monthly	Pump seal condition, filter condition, suction hose integrity
Fuel Flowmeter (calibrated)	Annual calibration by accredited lab	Calibration certificate with NABL reference, accuracy within ±0.5%
Bonding Cable (aircraft-to-bowser)	Monthly functional test	Clamp jaw condition, wire continuity (<1 Ohm), insulation intact

Equipment	Inspection Interval	Key Compliance Parameters
Bonding Cable (bowser-to-ground)	Monthly functional test	Same as above; ground stake security
Spill Kit	Monthly inspection	Absorbent pads qty (min 20), containment boom (5m), PPE (gloves, goggles), disposal bags
Water Detector Paste (Kolor Kut)	Per batch	Expiry date; stored sealed
Hydrometer (fuel density)	Annual calibration	Calibration certificate, range covers 0.680–0.740 kg/L
Fuel Sample Bottles (clear glass)	Per use	Clean, dry, no residue

6.4 Fire Safety Equipment

Equipment	Location	Type/Capacity	Inspection Interval
Fire Extinguisher	Hangar entrance (per aircraft)	CO2 5kg or DCP 9kg	6-monthly pressure check + Annual service
Fire Extinguisher	Fueling Point	DCP 9kg (minimum 2)	6-monthly
Fire Extinguisher	Operations Room	CO2 2kg (electrical safety)	6-monthly
Fire Extinguisher	Fuel Store	DCP 9kg	6-monthly
Fire Extinguisher	Hostel (per floor)	CO2 2kg	6-monthly
Fire Blanket	Hangar (near first aid station)	Fibreglass 1.2 × 1.2m	Annual inspection
Fire Alarm System	Hangar, Ops Room, Hostel, Fuel Store	Addressable smoke/heat detectors	Annual test by certified contractor
Evacuation Plans	All buildings	Laminated A3 charts at all exits	Annual review and update

6.5 Maintenance Tools

Tool	Calibration Required?	Interval	Notes
Torque Wrench Set (1/4", 3/8", 1/2" drive)	YES	Annual (NABL-accredited lab)	Range covers 0–200 in-lb; 0–150 ft-lb; calibration certificate stored in AMS

Tool	Calibration Required?	Interval	Notes
Standard Mechanic's Tool Kit	No	Visual inspection 6-monthly	Inventory count vs. shadow board; missing tools flagged before any maintenance
Safety Wire Pliers	No	6-monthly	Grip condition, spring tension
Borescope (engine cylinder inspection)	No	Annual	Fiber optic condition, light source, image quality check
Compression Tester	YES	Annual	Gauge calibration; adapter fitting condition
Pitot-Static Test Set	YES	Annual (NABL)	Leak rates, altitude and airspeed range, calibration certificate
Multimeter (electrical testing)	Recommended annual check	Annual	Calibration vs. known reference
Ohmmeter / Continuity Tester	Recommended	Annual	For bonding and ELT antenna resistance checks
Magneto Test Light	No	6-monthly	Bulb condition, lead insulation
Ground Power Unit (GPU)	Annual service	Annual	Voltage/frequency output check, cable condition

6.6 Communications Equipment

Equipment	Specification	License/Compliance
VHF Ground Station (Primary)	25W transceiver, frequency range 118–136.975 MHz	WPC license (Wireless Planning Commission); annual renewal; frequency allocation per DGCA/AAI
VHF Handheld (Backup)	5W handheld (Motorola/Sepura or similar)	WPC license (handheld category)
NOTAM Printer / Terminal	Computer with AAI NOTAM subscription	AAI account; updated at start of each flying day

6.7 Aerodrome Equipment

Equipment	Inspection	Regulatory Reference
Windsock (ICAO standard)	Monthly; replace if faded or torn	DGCA CAR Section 4, Series B — Windsock must indicate 15 kt wind at full extension

Equipment	Inspection	Regulatory Reference
PAPI System (if installed)	Monthly alignment check; annual glidepath verification	DGCA CAR Section 4; ICAO Annex 14 — 2.5°–3.5° glidepath depending on aircraft
Runway Threshold Lights	Monthly functional test; annual photometric check	ICAO Annex 14
FOD (Foreign Object Debris) Control Kit	Daily FOD check logged; kit inspected monthly	Runway/taxiway sweeper or personnel walk; any FOD found → logged as safety event

6.8 IT & Administrative Infrastructure

Equipment	Qty per Branch	Management in AMS
AMS Terminals (dedicated)	3 (Ops Room, Maintenance, Admin)	Asset record; software version; last security patch date
CCTV System	Hangar, runway, fuel point, hostel entrance, gates	Camera inventory; storage server; recording retention period (90 days)
Biometric Attendance Terminal	1 per entrance (Ops, Hangar, Hostel)	Device ID; firmware version; attendance data sync to AMS HR module
UPS System	1 per critical area (Ops Room, Server Room)	Capacity (KVA), battery installation date, load test log, next service
Generator (DG Set)	1 per branch (minimum 15 KVA)	Run log (hours), last service, fuel level, load test date
Server / NAS (if on-premises)	Optional (cloud is primary)	Hardware inventory, RAID status, backup log

7. DGCA COMPLIANCE CALENDAR — RECURRING OBLIGATIONS

7.1 Overview

The following table lists all recurring DGCA and regulatory compliance obligations that the AMS Compliance Calendar module tracks. All items generate automated alerts at 90, 60, 30, 14, 7, and 1 day(s) before due date. The responsible officer is the primary alert recipient; the escalation officer (CFI/Director) receives alerts at T-14 and below.

Item	Obligation	Regulatory Reference	Interval	Responsible	Escalation
CC-001	Aircraft Certificate of Airworthiness Renewal	Aircraft Rules 1937, Rule 50	Annual	CFI	Director
CC-002	Aircraft Certificate of Registration Renewal (if limited validity)	Aircraft Rules 1937	As per C of R	CFI	Director
CC-003	Aircraft Insurance Renewal	Aircraft Act 1934 / Policy terms	Annual (typical)	Director/ Admin	Director
CC-004	100-Hour Inspection (per aircraft)	Aircraft Rules 1937 / Cessna AMM	Every 100 flight hours	LAME / CFI	CFI
CC-005	Annual Inspection (per aircraft)	Aircraft Rules 1937, Rule 61	Every 12 calendar months	LAME	CFI
CC-006	Engine Oil Change (per aircraft)	Lycoming SSP-1003 / AMM	Every 50 hours or 4 months	LAME	CFI
CC-007	Spark Plug Rotation (per aircraft)	AMM	Every 50 hours	LAME	CFI
CC-008	Transponder Altimetry Check (per aircraft)	FAR 91.413 / DGCA equivalent	Every 24 months	LAME + Avionics shop	CFI
CC-009	Pitot-Static System Leak Check (per aircraft)	FAR 91.411 / DGCA equivalent	Every 24 months	LAME + certified shop	CFI
CC-010	ELT Annual Inspection (per aircraft)	FAR 91.207 / DGCA	Every 12 months	LAME	CFI
CC-011	ELT Battery Replacement	Per battery manufacturer / ADs	As per expiry or 50% cumulative tx time	LAME	CFI
CC-012	VHF Ground Station License Renewal (WPC)	Wireless Telegraphy Act 1933	Annual	Admin Officer	Director
CC-013	FTO Certificate Renewal	CAR Section 7, Series E	As per approval validity	Director	Director
CC-014	FI Rating Renewal (per instructor)	DGCA FI Rating Rules	Every 12 months	Instructor	CFI

Item	Obligation	Regulatory Reference	Interval	Responsible	Escalation
CC-015	CFI / DCFI Designation Renewal	DGCA requirement	As per DGCA approval	CFI	Director
CC-016	Student Medical (Class 2) Expiry Tracking	DGCA Regulations for Licensing	As per validity (1–5 years, age dependent)	Student + Admin	CFI
CC-017	Instructor Medical (Class 1) Expiry Tracking	DGCA Regulations for Licensing	12 months (under 40); 6 months (over 40)	Instructor	CFI
CC-018	Instructor Flight Currency (90-day per type)	CAR-FTO	Rolling 90 days	Auto-calculated by AMS	CFI
CC-019	FI Duty Time Review (Monthly)	CAR Section 7, Series E	Monthly audit	CFI	Director
CC-020	SMS Quarterly Review	DGCA CAP 8100	Quarterly	Safety Officer	Director
CC-021	Emergency Response Exercise	DGCA SMS Requirements	Annual	Safety Officer + CFI	Director
CC-022	DGCA Annual Returns Submission	CAR-FTO	Annual (31 March)	Director + Admin	Director
CC-023	Fire Extinguisher 6-Monthly Service (all)	NBC Act / Local Fire Safety	Every 6 months	Admin / Safety Off.	CFI
CC-024	Fire Alarm System Annual Test	NBC / Fire Safety	Annual	Admin	Director
CC-025	Torque Wrench Calibration (NABL)	Aircraft Rules / Best Practice	Annual	LAME	CFI
CC-026	Fuel Bowser Annual Service	AMM / Bowser manufacturer	Annual	LAME / Storekeeper	CFI
CC-027	Fuel Flowmeter Calibration	Weights & Measures Act	Annual	Storekeeper / LAME	CFI
CC-028	Aircraft Insurance Liability Cover Review	Company Policy	Annual	Director	Director
CC-029	Student SPL Renewal Tracking	DGCA Licensing Rules	Every 2 years (typical)	Student + Admin	CFI
CC-030	Propeller Overhaul (calendar-based)	Sensenich / McCauley SB	Per manufacturer (typically 72 months or TBO)	LAME	CFI
CC-031	GSTIN Return Filing (GSTR-1, GSTR-3B)	GST Act 2017	Monthly/ Quarterly	Accounts Officer	Director

Item	Obligation	Regulatory Reference	Interval	Responsible	Escalation
CC-032	TDS Return Filing (Form 24Q, 26Q)	Income Tax Act 1961	Quarterly	Accounts Officer	Director
CC-033	PF/ESI Returns	EPF Act / ESI Act	Monthly	HR Officer	Director
CC-034	RTI Response Deadline Tracking	RTI Act 2005	30 days from receipt	Admin / Legal Off.	Director
CC-035	AWS / Met Equipment Annual Calibration	IMD / WMO standards	Annual	Met Officer	CFI

8. RECOMMENDATIONS TABLE

Ref	Recommendation	Risk if Not Done	Priority
REC-01	Hire LAME Cat B1 at BOTH branches immediately	*No aircraft can be legally certified for flight without LAME sign-off (CRS). Aircraft are being flown without legal airworthiness certification.*	**CRITICAL — P1**
REC-02	Hire CFI at Pinjore immediately	*Pinjore cannot legally operate as an FTO without a CFI. All student training records from Pinjore may be invalidated.*	**CRITICAL — P1**
REC-03	Hire dedicated Flight Instructors (min 2 per branch)	*DGCA requires at least 1 FI per aircraft. CFI and DCFI are management roles and cannot exclusively fly line instruction.*	**HIGH — P2**
REC-04	Hire Ground Instructors at Karnal (CGI + min 2 GIs)	*Students at Karnal will not be eligible to appear for DGCA CPL theory examinations without qualified instruction.*	**HIGH — P2**
REC-05	Establish IMD Met Department at each branch	*Remote METAR from nearest aerodrome may not accurately reflect local conditions. A weather-related accident will likely be attributed to inadequate met services.*	**HIGH — P2 (Safety)**
REC-06	Appoint dedicated Safety Officer (SMS Focal Point)	*DGCA CAP 8100 mandates an SMS for all DGCA-approved organisations. Absence of a functional SMS is prima facie evidence of organisational negligence in the event of an accident.*	**HIGH — P2 (Regulatory)**
REC-07	Install ADS-B Out on all 6 aircraft	*ADS-B is increasingly required in all controlled airspace. Training aircraft near Chandigarh TMA will require ADS-B. Also enables AMS live tracking and overdue management.*	**MEDIUM-HIGH — P3**
REC-08	Procure FNPT-II or BITD Flight Simulator	*CPL syllabus allows up to 40% of required hours in approved simulators, providing instrument and emergency training in a zero-risk environment at lower cost per hour.*	**MEDIUM — P3**
REC-09	Implement biometric attendance at all branch entry points	*Manual attendance is subject to manipulation, and inaccurate duty time records undermine FI duty time compliance calculations in the AMS.*	**MEDIUM — P3**
REC-10	Establish robust IT infrastructure per branch	*Internet outages will cause AMS unavailability, reverting to manual operations. Power failures without UPS will cause data loss.*	**MEDIUM — P3**
REC-11	Develop and publish Operations Manual per DGCA CAR	*DGCA requires FTOs to have an approved Operations Manual. Without it, DGCA cannot renew FTO approval.*	**HIGH — P2 (Regulatory)**
REC-12	Procure dedicated Fuel Quality Testing Kit per branch	*There is currently no documented procedure or equipment for fuel quality acceptance testing — a safety-critical gap that could allow contaminated fuel to reach aircraft engines.*	**CRITICAL — P1 (Safety)**

9. METEOROLOGICAL DEPARTMENT — FULL PLAN

9.1 Current Status and Regulatory Context

Neither HICA Karnal nor HICA Pinjore currently has a formal meteorological department. This represents a significant safety and compliance gap. DGCA regulations for FTOs require that adequate weather information be available to pilots for flight planning and pre-flight briefing. Where no aerodrome met office exists, the FTO is required to have a documented procedure for obtaining, verifying, and disseminating weather information.

9.2 Scenario B — Interim Arrangement (No IMD Dept.) — Current Status

Procedure (to be formalised and documented in Operations Manual)

- At 0600 daily: DCFI logs into AMS Met Module; system auto-fetches METAR for: VIKR/Karnal (or nearest AWS); VOCX Chandigarh (for Pinjore branch)
- DCFI conducts local observation: anemometer reading (if installed) or streamer/smoke observation for wind direction and approximate speed; horizon visibility estimate; cloud base estimate (if ceilometer available) or sky observation
- Enters local observations in AMS with mandatory fields: observation time, observer name, visibility (km), cloud (amount and estimated base), wind (direction and speed), pressure (barometer reading)
- AMS generates a formatted local weather summary combining external METAR and local observation with mandatory disclaimer: 'Not an official meteorological service. Data for pilot information only. PIC is responsible for weather assessment.'
- CFI reviews summary and signs off Go / Weather Hold for the day (logged in AMS with timestamp)
- Any deterioration during the day: DCFI or dispatcher updates local observation; AMS triggers re-evaluation of Go/No-Go for any sortie in progress or planned

9.3 Scenario A — Recommended: Establish IMD Met Dept.

9.3.1 Staffing Requirements

Position	Qualification	Functions
Meteorological Officer (Grade II)	IATA/IMD Met Course + DGCA authorization; ideally B.Sc. Meteorology/Physics	Lead forecaster; issues TAFs; pilot briefings; maintains instruments; liaises with Regional Met Centre
Met Observer	IMD Observer Training Course (4 weeks)	Hourly/half-hourly surface observations; METAR encoding; SPECI on significant change; maintains obs log

9.3.2 Meteorological Equipment List

Equipment	Specification	Qty per Branch	Calibration Interval
Automatic Weather Station (AWS)	Integrated sensors: temp, dew point, pressure, wind (2 min average + gusts), RVR estimate	1	Annual by IMD/NABL
Cup Anemometer + Wind Vane	Standalone backup; calibrated; height per ICAO Annex 3 (10m AGL)	1	Annual
Ceilometer (Cloud Base)	Laser ceilometer (Vaisala CL31 or equivalent); 0–7,700m range	1	Annual by manufacturer
Barometer (aneroid)	Precision aneroid; sea-level pressure reduction capability	1	Annual (compare vs. calibrated standard)
Thermometer / Thermograph	Dry/wet bulb (Stevenson screen) for dewpoint/humidity	1 set	6-monthly check
Visibility Sensor (PWD)	Present Weather Detector (Vaisala FD12P or similar); 10m–75km range	1	Annual
Hygrometer (relative humidity)	Capacitive sensor; ±2% accuracy	1	Annual
Met Display Terminal	AWS data display PC; METAR generation software	1	Software updates as released
Barograph / Pressure Recorder	Analogue backup for pressure trend	1	6-monthly
Upper Wind Report Facility	Access to regional radiosonde data via internet/IMD API	1 (software)	N/A

9.3.3 AMS Integration (Scenario A)

- AWS data streams into AMS Met Module via MQTT protocol at 1-minute intervals
- AMS auto-generates METAR at HH+20 and HH+50 every hour using template populated from AWS data plus manual observer inputs (sky condition, present weather, remarks)
- SPECI auto-triggered when: visibility drops below 5km, 3km, 1.5km; ceiling drops below 1,500ft, 1,000ft, 500ft; wind gusts exceed 25kt; sudden onset of thunderstorm or freezing precipitation
- TAF issued 4 times daily (0000, 0600, 1200, 1800 UTC) by Met Officer; stored in AMS; pushed to pilot briefing board
- All METARs, SPECIs, and TAFs stored in AMS with immutable timestamp; constitute legal weather record for each sortie.

10. USER ONBOARDING FLOW

10.1 New Staff Member Onboarding

When the HR Officer (or Super Admin) creates a new employee record, the AMS triggers a structured onboarding workflow:

1. HR Officer enters basic details in Employee Master (name, role, contact, start date)
2. System sends a welcome email with temporary login credentials and prompts user to set a strong password and enable MFA
3. Onboarding Checklist auto-generated based on role. Examples:
   - For a Flight Instructor: Upload CPL/ATPL license scan, FI Rating scan, Class 1 Medical scan, sign HICA Operations Manual receipt, complete system orientation module, complete SMS awareness briefing
   - For a LAME: Upload LAME License (B1/B2), relevant type endorsements, Class 2 Medical, complete AMS maintenance module orientation
4. Checklist completion tracked in HR module; user cannot access full system until mandatory items are complete
5. HR Officer and CFI notified when onboarding is 100% complete
6. Compliance Calendar automatically populated with this individual's license/rating/medical expiry dates

10.2 New Aircraft Onboarding

7. AME/CFI initiates 'Add Aircraft' wizard from Fleet Module
8. Step 1: Select type (Cessna 152 / Cessna 172 / Other) — pre-built template loads
9. Step 2: Enter registration, serial number, year of manufacture
10. Step 3: Enter current TTAF, engine TSN/TSOH, propeller TSN/TSOH
11. Step 4: Enter document dates: C of A (issue/expiry), C of R, Insurance
12. Step 5: Avionics register — system presents template list; AME confirms each instrument and enters S/N and last calibration date
13. Step 6: AD Compliance — system queries FAA AD database using aircraft S/N; generates AD list; AME reviews and enters compliance status for each applicable AD
14. Step 7: Initial maintenance status — AME enters last 50-hour, 100-hour, Annual dates and TTAF at those checks
15. Step 8: First CRS entry — LAME issues initial CRS certifying aircraft is airworthy as at introduction date
16. Step 9: Aircraft now available in Fleet Module, Go/No-Go Engine, and AI Rostering Engine

10.3 New Branch Onboarding

17. Super Admin initiates 'Add Branch' wizard from System Configuration
18. Enter: Branch name, ICAO code, physical address, GPS coordinates, timezone
19. Aerodrome details: runway designation, magnetic variation, elevation, circuit height, circuit direction
20. Emergency contacts for this branch: local fire brigade, ambulance, police, nearest hospital, DGCA Regional Office
21. Assign: Branch Director, CFI (required), DCFI
22. Copy base configuration from existing branch (optional — saves 80% of setup time for a clone branch)
23. Aircraft and personnel already in the system can be re-assigned or new ones added
24. Compliance Calendar auto-populated for the new branch based on initial configuration
25. Go-live verification: System self-check flags any critical gaps (e.g., no LAME assigned, no CRS for any aircraft)

10.4 New FTO / Club Onboarding (SaaS White-Label)

This flow targets a brand-new customer organization (not HICA-specific):

1. Self-service registration at www.aviams.in (or white-label URL): Organization name, DGCA FTO number, contact person
2. Automated account creation; Super Admin credentials sent by email with MFA setup required
3. Onboarding Wizard launched: 8 steps (as described in Module 19, Section 3)
4. Data import: Aircraft master import (Excel template), Personnel import (Excel template), Student import (Excel template)
5. System validation: All imported data checked against AMS rules; import errors highlighted with fix guidance
6. Configuration: DGCA compliance calendar pre-populated from standard obligation library; FTO reviews and adjusts dates
7. Integration setup: Guided setup for Garmin Pilot, biometric device, WhatsApp, SMS
8. Trial and verification flight: Test sortie scheduled and executed in system with test data
9. Go-Live: All production data replaced; live operations commence

Target onboarding duration: 48 hours from registration to first live sortie recording. Supported by online documentation, video walkthroughs, and live chat support.